LOW-LYING ZEROS OF NEWFORMS IN THE DEPTH ASPECT AT THE PRIME 2, SPLIT BY ROOT NUMBER

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ABSTRACT. We set an even weight $k \geq 2$ and study newforms at increasingly large levels $N = 2^n$ for both even and odd exponents n. Without assumption of any Riemann Hypothesis, we show that the natural one-level density of the full newspace agrees with orthogonal symmetry for even Schwartz tests with compact Fourier support in $(-3/2, 3/2)$, with error $O_{k,\phi}(\log(k^2N)^{-1})$. The proof of this result requires the use of the fact that all elliptic terms within the newspace trace cancel completely, except for when $(N/4) \mid t^2 - 4m$. Under the Generalized Riemann Hypothesis for primitive Dirichlet L-functions, the root number families $+1$ and -1 have SO(even) and SO(odd) densities, respectively, for support in $(-4/3, 4/3)$, along with the same error $O_{k,\phi}(\log(k^2N)^{-1})$. Additionally, we determine the logarithmic mean, $1/2$, of the associated murmuration density as well as its oscillations of order $y^{1/4}$.

1. INTRODUCTION

The predicted densities SO(even) and SO(odd) agree on Fourier support in the interval $[-1, 1]$. To distinguish the two root-number subfamilies, we therefore need support beyond 1 and estimates uniform in growing Hecke indices.

1.1. **Results.** We fix an even integer $k \geq 2$, set $N = 2^n$ and $L = \log(k^2N)$, and let $H_k^{\text{new}}(N)$ be a Hecke eigenbasis of $S_k^{\text{new}}(N)$ with trivial nebentypus. We write $d_N = |H_k^{\text{new}}(N)|$ and $\varepsilon_f = i^k w_f$, where w_f is the Fricke eigenvalue. For an even Schwartz function ϕ with compactly supported Fourier transform, we define

$$D(f;\phi) = \sum_{\rho_f} \phi\left(\frac{\gamma_f L}{2\pi}\right), \qquad \rho_f = \tfrac{1}{2} + i\gamma_f,$$

where each nontrivial zero of $L(s, f)$ is counted with its multiplicity, and we let

$$U_N(\phi) = \frac{1}{d_N} \sum_f D(f;\phi), \qquad S_N(\phi) = \frac{1}{d_N} \sum_f \varepsilon_f D(f;\phi).$$

Here f runs over $H_k^{\text{new}}(N)$, and we let $D_N^{\pm}(\phi)$ denote the average over the forms with $\varepsilon_f = \pm 1$. Each newform receives equal weight. Because we do not assume modular GRH, γ_f may be complex, so we evaluate ϕ using its entire extension (Lemma 4.5). Nonnegativity of ϕ on $\mathbb{R}$ therefore need not make the terms in $D(f;\phi)$ nonnegative. These density identities alone do not justify using nonnegative zero sums to bound orders of vanishing at the central point or to show that a positive proportion of the central values are nonzero.

Theorem 1.1. *Let $k \geq 2$ be an even integer and let ϕ be an even Schwartz function with* $\operatorname{supp} \widehat{\phi}$ *compactly contained in $(-3/2, 3/2)$. Then, unconditionally, for $N = 2^n$ with $n \geq 8$, through even and odd n,*

$$U_N(\phi) = \widehat{\phi}(0) + \frac{\phi(0)}{2} + O_{k,\phi}\big(L^{-1}\big) = \int_{\mathbb{R}} \phi(x) W(\mathrm{O})(x)\, dx + O_{k,\phi}\big(L^{-1}\big).$$

Theorem 1.2. *Assume the Riemann hypothesis for every primitive Dirichlet L-function, including ζ. Let $k \geq 2$ be an even integer and let ϕ be an even Schwartz function with $\operatorname{supp}\widehat{\phi}$ compactly contained in $(-4/3, 4/3)$. Then, for $N = 2^n$ with $n \geq 7$,*

$$S_N(\phi) = -\frac{1}{2}\int_{|\xi|>1} \widehat{\phi}(\xi)\, d\xi + O_{k,\phi}\big(L^{-1}\big)\,.$$

No hypothesis is made on the zeros of any modular L-function.

Corollary 1.3. *Under the Riemann hypothesis for primitive Dirichlet L-functions alone, for even Schwartz ϕ with $\operatorname{supp}\widehat{\phi}$ compactly contained in $(-4/3, 4/3)$ and $n \geq 8$,*

$$D_N^+(\phi) \to \int_{\mathbb{R}} \phi\, W(\mathrm{SO(even)}), \qquad D_N^-(\phi) \to \int_{\mathbb{R}} \phi\, W(\mathrm{SO(odd)}),$$

each with rate $O_{k,\phi}(L^{-1})$.

1.2. **Prior work and proof strategy.** For restricted Fourier support and without assumption of any Riemann Hypothesis, Shin–Templier [ST16, Examples 9.10 and 9.14, Theorem 11.5] found natural one-level densities in the weight aspect at level one. Under the Generalized Riemann Hypothesis for modular, symmetric-square, and Dirichlet L-functions, Barrett et al. [BBD+17, Theorem 1.3] obtain natural newform averages with support $(-2, 2)$ at arbitrary levels. However, their result concerns the combined family, and does not split the family by root number at general level [BBD+17, Remark 1.4].

For a fixed odd prime, Burrin et al. [BKL+26, Theorem 1.1] prove an unconditional murmuration theorem at odd conductor exponents. Tomczak [Tom26, Theorem 1.1(i)] treats even exponents under the Generalized Riemann Hypothesis for all Dirichlet L-functions. Both papers average over a fixed range of prime-to-conductor ratios. Under the Generalized Riemann Hypothesis, Kroesche [Kro23, Theorem 1.2.7] obtains support beyond 1 for selected families with a fixed root number. Here, every newform at level 2^n receives equal weight, instead of using Petersson weights like Kroesche.

In order to remove the oldform contributions for the unsigned average, we apply a second difference to Popa's trace formula [Pop18, Theorem 3]. Stabilization of the local projective-root counts forces $(N/4) \mid t^2 - 4m$. Counting the pairs of primes and trace indices that satisfy this restriction then gives support below $3/2$. For the signed trace, Dirichlet GRH gives a uniform $B^{3/4+\epsilon}$ error in prime averaging. This estimate works over the growing trace-index range and gives support below $4/3$. The main term of the signed one-level density comes from the logarithmic mean of the murmuration density (Section 6). Support beyond $3/2$ is known for harmonic prime-level averages, as per Devin et al. [DFS25, Theorem 1.1].

1.3. **Notation.** Throughout, P denotes an odd prime. The Hecke eigenvalues are Deligne-normalized. This is done so that $|\lambda_f(p)| \leq 2$ for $p \nmid N$, and $\lambda_f(2) = 0$ since $4 \mid N$. T_m, the classical Hecke operator, acts with eigenvalue $m^{(k-1)/2}\lambda_f(m)$.

For a negative discriminant $D = d\ell^2$ with fundamental part d we write: $\psi_D(m) = \left(\frac{d}{m/(m,\ell)}\right)$ for the Kronecker symbol, $L(s, \psi_D) = \sum_{m\geq 1} \psi_D(m) m^{-s}$, and $H(D)$ for the Hurwitz class number of $|D|$, so that $H(D) = \pi^{-1} L(1, \psi_D)\sqrt{|D|}$ for $D < 0$. We set $H(0) = -1/12$ and $H = 0$ at inadmissible arguments, and h_0 signifies the weighted primitive class number.

We use $\widehat{\phi}(\xi) = \int_{\mathbb{R}} \phi(x) e^{-2\pi i x\xi}\, dx$. The Katz–Sarnak densities are

$$W(\mathrm{O})(x) = 1 + \tfrac{1}{2}\delta_0(x), \quad W(\mathrm{SO(even)})(x) = 1 + \frac{\sin 2\pi x}{2\pi x}, \quad W(\mathrm{SO(odd)})(x) = 1 - \frac{\sin 2\pi x}{2\pi x} + \delta_0(x),$$

so that $\int \phi\, W(\mathrm{SO(odd)}) - \int \phi\, W(\mathrm{SO(even)}) = \int_{|\xi|>1} \widehat{\phi}$.

Constants only depend on their subscripts, with C_{abs} absolute. Dependence on a test function includes its support and finitely many Schwartz seminorms. Unless otherwise indicated, the implied constants remain independent of the depth, Hecke index, and endpoints. However, they do not necessarily remain uniform as the support approaches a theorem's limiting value.

2. The trace identities

Once again, let $k \geq 2$ be even, $N = 2^n$, and $m \geq 1$ odd. We write

$$F_r(m) = \operatorname{tr}\big(T_m, S_k(\Gamma_0(2^r))\big), \qquad F_r^{\mathrm{new}}(m) = \operatorname{tr}\big(T_m, S_k^{\mathrm{new}}(2^r)\big),$$

and $d_N = \dim S_k^{\mathrm{new}}(N)$.

2.1. The full-space formula.

We use Popa's trace formula for trivial character and even weight [Pop18, Theorem 3 and (2.18)].

Let $\psi(Q) = Q\prod_{p\mid Q}(1+p^{-1})$, with $\psi(1)=1$, and let $p_{k-2}(t,m)$ be the coefficient of z^{k-2} in the generating function

$$\frac{1}{1-tz+mz^2} = \sum_{j\geq 0} p_j(t,m) z^j. \tag{1}$$

For $a \geq 0$, write $H_+(a) = H(-a)$. For $v \mid Q$ with $v^2 \mid t^2-4m$, define

$$\begin{aligned} B_Q(v,t,m) &= \frac{\psi(Q)}{\psi(Q/v)} \#\{\alpha \in (\mathbb{Z}/Q\mathbb{Z})^\times : \alpha^2 - t\alpha + m \equiv 0 \pmod{Qv}\}, \\ C_Q(v,t,m) &= \sum_{a\mid v} \mu(a) B_Q(v/a,t,m), \\ \Phi_Q(a,d) &= \sum_{\substack{rs=Q \\ (r,s)\mid a-d}} \varphi((r,s)), \\ \sigma_{1,Q}(m) &= \sum_{\substack{d\mid m \\ (d,Q)=1}} \frac{m}{d}. \end{aligned} \tag{2}$$

Our initial divisibility conditions cause the first set to be well defined modulo Q [Pop18, footnote 4].

Theorem 2.1 ([Pop18], Theorem 3, specialized)**.** *For every integer $Q \geq 1$, every even $k \geq 2$ and every positive Hecke index m, with classically normalized Hecke operators,*

$$\begin{aligned} \operatorname{tr}(T_m, S_k(\Gamma_0(Q))) = &-\frac{1}{2}\sum_{t^2\leq 4m} p_{k-2}(t,m) \sum_{\substack{v\mid Q \\ v^2 \mid t^2-4m}} H_+\left(\frac{4m-t^2}{v^2}\right) C_Q(v,t,m) \\ &-\frac{1}{2}\sum_{ad=m} \min(a,d)^{k-1}\Phi_Q(a,d) + \mathbf{1}_{k=2}\sigma_{1,Q}(m). \end{aligned} \tag{3}$$

For $(m,Q)=1$, the terms with $t^2 = 4m$ give the scalar contribution

$$\mathcal{I}_Q(m) = \frac{k-1}{12}\psi(Q) m^{k/2-1}\mathbf{1}_{m=\square} \tag{4}$$

by [Pop18, Remark 3.8].

Let $h(D)$ count primitive positive definite classes of discriminant $D<0$, $w(D)$ be their count for automorphisms, and set $h_0(D) = 2h(D)/w(D)$. Then [Pop18, (2.11)] and the following lemma

give

$$H(D)=\sum_{u^2\mid D} h_0(D/u^2),\qquad 0\le h_0(D)\le H(D)\le C_\delta^{\mathrm{cl}}|D|^{1/2+\delta}\quad (D<0,\ \delta>0). \tag{5}$$

Lemma 2.2 (A uniform class-number bound)**.** *For every $\delta>0$ there exists a constant $C_\delta^{\mathrm{cl}}>0$ depending only on δ such that every negative discriminant D satisfies*

$$|L(1,\psi_D)|\le C_\delta^{\mathrm{cl}}|D|^{\delta},\qquad H(D)\le C_\delta^{\mathrm{cl}}|D|^{1/2+\delta}. \tag{6}$$

Proof. Write $D=df^2$ with fundamental discriminant $d<0$. For $p^e\|f$, the definition gives $\psi_D(p^b)=\chi_d(p)^{\max(b-e,0)}$. First for $\operatorname{Re}s>1$ and then by continuation, summing the geometric series at each prime yields

$$L(s,\psi_D)=L(s,\chi_d)\prod_{p^e\|f}\left(1+(1-\chi_d(p))\sum_{j=1}^{e}p^{-js}\right). \tag{7}$$

The nonprincipal character χ_d has partial sums which are bounded by $|d|$. As a result, we can split at $|d|$ and apply partial summation to get

$$|L(1,\chi_d)|\le 2+\log|d|. \tag{8}$$

Let $T_\delta=\max(3,4/\delta)$ and $A_\delta=\prod_{p<T_\delta}(1+2/(p-1))$. If $p\ge T_\delta$, then $\log(1+2/(p-1))\le 4/p\le\delta\log p$. Consequently, the finite factor in (7) at $s=1$ must be at most $A_\delta f^\delta$. Because $f\le|D|^{1/2}$ and $2+\log|d|\le(2+2/(\mathrm{e}\delta))|D|^{\delta/2}$, the first bound holds, with $C_\delta^{\mathrm{cl}}=A_\delta(2+2/(\mathrm{e}\delta))$. The identity $H(D)=\sqrt{|D|}L(1,\psi_D)/\pi$ [BKL+26, (2.4)] gives the second bound with the same constant, and it is valid also for imprimitive $D<0$. □

We can substitute (5) into (3) and then use $\sum_{v\mid a}C_Q(v,t,m)=B_Q(a,t,m)$ to express the strict elliptic part in terms of primitive class numbers [Pop18, (2.16)]:

$$\mathcal{E}_Q(m)=-\frac12\sum_{t^2<4m}p_{k-2}(t,m)\sum_{u^2\mid t^2-4m}h_0\!\left(\frac{t^2-4m}{u^2}\right)B_Q((Q,u),t,m). \tag{9}$$

2.2. **The oldform decomposition and the newspace trace.** The oldspace decomposition [BBD+17, §2.1, (2.3)–(2.4)] for positive level, even weight, and trivial nebentypus is

$$S_k(\Gamma_0(Q))=\bigoplus_{LM=Q}\ \bigoplus_{f\in H_k^{\mathrm{new}}(M)}\operatorname{span}\{f_{|\ell}:\ell\mid L\},\qquad f_{|\ell}(z)=\ell^{k/2}f(\ell z). \tag{10}$$

The forms $f_{|\ell}$ in every block are linearly independent and satisfy $T_m(f_{|\ell})=(T_mf)_{|\ell}$ by [BBD+17, (2.5)] for $(m,Q)=1$.

Lemma 2.3 (The second-difference formula)**.** *Let $k\ge2$ and m be even and odd, respectively. Then*

$$F_n(m)=\sum_{a=0}^{n}(n-a+1)F_a^{\mathrm{new}}(m),\qquad F_n^{\mathrm{new}}(m)=F_n(m)-2F_{n-1}(m)+F_{n-2}(m)\quad (n\ge2). \tag{11}$$

Proof. By taking traces in (10), each level-2^a block contributes its eigenvalue exactly $n-a+1$ times. The second identity is achieved by two differences. □

2.3. **The newspace formula.** Write $D = t^2 - 4m$ and for $r \geq 2$, set

$$\Xi_r(t,m) = \sum_{j=0}^{r} H_+\left(\frac{4m-t^2}{4^j}\right) C_{2^r}(2^j,t,m), \qquad K_n = \Xi_n - 2\Xi_{n-1} + \Xi_{n-2}, \tag{12}$$

$$G_n(a,d) = \Phi_{2^n}(a,d) - 2\Phi_{2^{n-1}}(a,d) + \Phi_{2^{n-2}}(a,d). \tag{13}$$

By the class-number convention, contributions only come from j with $4^j \mid D$.

Proposition 2.4 (Untwisted newspace trace formula)**.** *For even $k \geq 2$, odd $m \geq 1$, and $n \geq 8$, with $N = 2^n$,*

$$F_n^{\mathrm{new}}(m) = \mathbf{1}_{m=\square}\frac{(k-1)N}{32}m^{k/2-1} - \frac{1}{2}\sum_{t^2<4m} p_{k-2}(t,m)K_n(t,m) - \frac{1}{2}\sum_{ad=m}\min(a,d)^{k-1}G_n(a,d). \tag{14}$$

Proof. Apply (11) to (3). The elliptic and cusp terms give K_n and G_n. The weight-two term cancels because $\sigma_{1,2^r}(m) = \sigma_1(m)$ at all three levels. From (4) and $\psi(2^r) = 3\cdot 2^{r-1}$, we get the scalar coefficient. □

To normalize the trace, set

$$V_N(m) = \frac{1}{d_N}\sum_{f\in H_k^{\mathrm{new}}(N)}\lambda_f(m) = \frac{F_n^{\mathrm{new}}(m)}{d_N m^{(k-1)/2}}.$$

Let $E_N(m)$ and $C_N(m)$ be the elliptic and cusp terms normalized by $d_N m^{(k-1)/2}$. For $n \geq 8$, lemma 2.5 gives

$$\frac{(k-1)N}{64} \leq d_N \leq \frac{(k-1)N}{32}, \qquad c_N := \frac{(k-1)N}{32d_N} = \begin{cases} 1, & n \text{ odd}, \\ \left(1 - 4/((k-1)\sqrt{N})\right)^{-1}, & n \text{ even}, \end{cases} \tag{15}$$

$$|c_N - 1| \leq \frac{8}{k-1}N^{-1/2}, \qquad 1 \leq c_N \leq 2 \quad (n \geq 8).$$

Hence, for all odd m,

$$V_N(m) = c_N m^{-1/2}\mathbf{1}_{m=\square} + E_N(m) + C_N(m). \tag{16}$$

Lemma 2.5 (Dimension at a power of two)**.** *For even $k \geq 2$,*

$$\dim S_k^{\mathrm{new}}(2^a) = (k-1)2^{a-5} \qquad (a \text{ odd},\ a \geq 5),$$
$$\dim S_k^{\mathrm{new}}(2^n) = (k-1)2^{n-5} - 2^{n/2-3} \qquad (n \text{ even},\ n \geq 4).$$

In particular $d_N \geq (k-1)N/64$ for $n \geq 8$, and $\left|(k-1)N/(32d_N) - 1\right| \leq 8(k-1)^{-1}N^{-1/2}$.

Proof. For $n \geq 4$, Martin's formula [Mar05, Theorem 1 and Definition 1′] has vanishing elliptic and $\mu(N)$ terms and local factors

$$s_0^{\#}(2^n) = \frac{3}{8}, \qquad \nu_\infty^{\#}(2^n) = \begin{cases} 0, & n \text{ odd}, \\ 2^{n/2-2}, & n \text{ even}. \end{cases} \tag{17}$$

Substituting into $d_N = (k-1)Ns_0^{\#}(N)/12 - \nu_\infty^{\#}(N)/2$ gives the identities, while $4/((k-1)\sqrt{N}) \leq 1/4$ for $n \geq 8$ gives the bounds. □

3. Local cancellation and a uniform unsigned trace bound

Throughout, $m \geq 1$ is odd and $N = 2^n$ with $n \geq 8$. The level in (9) only occurs in B_Q, and we determine when its second difference vanishes.

The local fixed-line count. Let

$$M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \qquad \det M = m \text{ odd}, \quad \operatorname{tr} M = t, \qquad Q_M(x,y) = cx^2 + (d-a)xy - by^2, \tag{18}$$

let $u > 0$ be the content of Q_M, and let $D = t^2 - 4m < 0$ be its discriminant. Popa's (2.17), Lemma 4.1, and the proof of that lemma interpret $B_{2^h}\big((2^h, u), t, m\big)$ as the number $\varrho_h(M)$ of projective roots of Q_M modulo 2^h.

Cosets of $\Gamma_0(2^h)$ actually correspond to primitive bottom rows up to a unit and $(XMX^{-1})_{21} = Q_M(d_X, -c_X)$. If $(XMX^{-1})_{21}$ vanishes modulo 2^h, both diagonal entries of XMX^{-1} are units because $\det M$ is odd.

All of the primitive forms $[A, B, C]$ of discriminant D/u^2 in (9) have the representative

$$M = \begin{pmatrix} (t - uB)/2 & -uC \\ uA & (t + uB)/2 \end{pmatrix}. \tag{19}$$

Its entries must be integers because of the congruence $t^2 \equiv u^2 B^2 \pmod 4$. Also, its determinant is m, trace is t, and associated form is $u[A, B, C]$.

Set $e = v_2(u)$. Let the number of projective roots of the form Q_M/u modulo 2^j be denoted by R_j. With y being even, the charts $(x, 1)$ and $(1, y)$ give us

$$\varrho_h(M) = \begin{cases} 3 \cdot 2^{h-1}, & 1 \le h \le e, \\ 2^e R_{h-e}, & h > e. \end{cases} \tag{20}$$

When $e < h$, reducing by 2^e has 2^e lifts for every point, and odd content does not impact the roots.

We write $Ax^2 + Bxy + Cy^2$ for the primitive form. We have two cases. If B is odd, then the derivative in either chart must be odd, which means that every root has precisely one lift per step and hence $R_j = R_1 \le 3$ for $j \ge 1$.

However, if B is even then at least one of A and C must be odd. We can assume A is odd after interchanging coordinates. Then, we have no roots with y even, so completing the square gives us

$$A(Ax^2 + Bx + C) = (Ax + B/2)^2 - \Delta, \qquad \Delta = \frac{B^2 - 4AC}{4} \neq 0. \tag{21}$$

In this case, R_j counts square roots of Δ modulo 2^j, because the substitution $x \mapsto Ax + B/2$ is bijective modulo 2^j.

Set $v = v_2(\Delta)$. There are exactly $2^{\lfloor j/2 \rfloor}$ roots if $j \le v$. There are none if $j > v$ and v is odd. If $j > v = 2a$, their number is 2^a multiplied by the number of odd roots of $\Delta/2^{2a}$ modulo 2^{j-2a}. Any two roots satisfy $2^b \mid (z - z_0)(z + z_0)$ for an odd square congruence modulo 2^b. The valuation of one factor is 1, so for $b \ge 3$ all roots are congruent to $\pm z_0 \pmod{2^{b-1}}$. Consequently, there is a maximum of four roots. There are four roots of an odd unit for all $b \ge 3 \iff$ the odd unit is $1 \pmod 8$. Starting at $b = 3$, we pair z with $z + 2^{b-1}$. Their squares actually differ by $2^b \pmod{2^{b+1}}$, which means exactly one member has two lifts. Combined with the immediate cases where $b = 1, 2$, this proves that

$$R_j \le 4 \cdot 2^{j/2} \quad (j \ge 1), \qquad R_j \text{ is constant for } j \ge v_2(\Delta) + 3 \quad \text{in the even-}B\text{ case.} \tag{22}$$

It follows, also when $h \le e$, from (20)–(22) that

$$0 \le \varrho_h(M) \le 4 \cdot 2^{(h+e)/2}, \qquad \big|\varrho_n(M) - 2\varrho_{n-1}(M) + \varrho_{n-2}(M)\big| \le 16 \cdot 2^{(n+e)/2}. \tag{23}$$

Lemma 3.1 (Local cancellation)**.** *Let m be odd, $t^2 < 4m$, $D = t^2 - 4m$, and M be a matrix as in (18) with content u, and set $e = v_2(u)$. Then*

$$\varrho_n(M) - 2\varrho_{n-1}(M) + \varrho_{n-2}(M) = 0 \quad \textit{unless} \quad v_2(D) \ge n + e - 2. \tag{24}$$

Proof. The trivial case is if $e \geq n-2$, then $v_2(D) \geq 2e \geq n+e-2$ and there is nothing to prove. Otherwise e is less than all three levels, so (20) reduces the claim to stabilization of R_j. We have two cases. If B is odd then these counts are simply constant. If B is even, then the failure of the condition in (24) implies

$$n-2-e \geq v_2(D) - 2e + 1 = v_2(\Delta) + 3. \tag{25}$$

Hence all three counts lie within the stable range of (22). □

Corollary 3.2. *With $K_n(t,m)$ the second difference of the local elliptic weights,*

$$K_n(t,m) = 0 \quad \textit{unless} \quad N/4 \mid t^2 - 4m. \tag{26}$$

Size bounds for the three contributions. For $u = 2^e v$ with v odd, (5), (23) and Lemma 3.1 give

$$h_0(D/u^2)\left|\varrho_n - 2\varrho_{n-1} + \varrho_{n-2}\right| \leq 16 C_\delta^{\mathrm{cl}} N^{1/2} |D|^{1/2+\delta} 2^{-e(1/2+2\delta)} v^{-1-2\delta} \mathbf{1}_{N/4|D}. \tag{27}$$

For every fixed $\delta > 0$ both extended sums converge, therefore

$$\sum_{u^2|D} h_0(D/u^2)\left|\varrho_n - 2\varrho_{n-1} + \varrho_{n-2}\right| \leq \frac{16 C_\delta^{\mathrm{cl}} \zeta(1+2\delta)}{1 - 2^{-1/2-2\delta}} N^{1/2} |D|^{1/2+\delta} \mathbf{1}_{N/4|D}. \tag{28}$$

The roots of $X^2 - tX + m$ have modulus $\sqrt{m}$ for $t^2 < 4m$, so we can expand the generating function to get

$$|p_{k-2}(t,m)| \leq (k-1) m^{(k-2)/2}. \tag{29}$$

For the cusp contribution, direct summation of Euler totients gives us

$$\sum_{j=0}^{h} \varphi\big(2^{\min(j,h-j)}\big) = \begin{cases} 3 \cdot 2^{h/2-1}, & h \geq 2 \text{ even}, \\ 2^{(h+1)/2}, & h \text{ odd}, \end{cases} \qquad 0 \leq \Phi_{2^h}(a,d) \leq 2 \cdot 2^{h/2}, \tag{30}$$

whose second difference has an absolute value of at most $6N^{1/2}$. Normalizing by (15) yields

$$|C_N(m)| \leq \frac{192}{k-1} N^{-1/2} \sum_{ad=m} \left(\frac{\min(a,d)}{\sqrt{m}}\right)^{k-1}, \tag{31}$$

and specifically with $K_k = 576/(k-1)$,

$$|C_N(P)| \leq K_k N^{-1/2} P^{-(k-1)/2}, \qquad |C_N(P^2)| \leq K_k N^{-1/2}. \tag{32}$$

The surviving indices and the uniform bound. In order to count the indices allowed by Corollary 3.2, let

$$\begin{gathered} Q = \frac{N}{4}, \qquad q = \frac{Q}{4} = \frac{N}{16}, \\ T_N(m) = \#\{t \in \mathbb{Z} : t^2 < 4m,\ Q \mid 4m - t^2\}, \end{gathered} \tag{33}$$

and set

$$K_\delta = \frac{512 \cdot 4^{1/2+\delta} C_\delta^{\mathrm{cl}} \zeta(1+2\delta)}{1 - 2^{-1/2-2\delta}}. \tag{34}$$

Combining (28), (29), and the dimension bound (15) with $|D| \leq 4m$ gives

$$|E_N(m)| \leq K_\delta N^{-1/2} m^\delta T_N(m) \qquad (m \geq 1 \text{ odd},\ \delta > 0). \tag{35}$$

Theorem 3.3 (Uniform unsigned trace bound)**.** *Let $k \geq 2$ be even, $N = 2^n$ with $n \geq 8$, and $\delta > 0$. Set $C_{k,\delta} = K_\delta + 576/(k-1)$. Then for every odd $m \geq 1$,*

$$\left| V_N(m) - \frac{\mathbf{1}_{m=\square}}{\sqrt{m}} \right| \leq C_{k,\delta} N^{-1/2} \left[m^\delta T_N(m) + \sum_{ad=m} \left(\frac{\min(a,d)}{\max(a,d)} \right)^{(k-1)/2} \right]. \tag{36}$$

In particular, for every odd prime P,

$$|V_N(P)| \leq C_{k,\delta} N^{-1/2} \left[P^\delta T_N(P) + P^{-(k-1)/2} \right]. \tag{37}$$

Proof. Insert (35), (31) and (15) into (16), using $\min(a,d)/\sqrt{m} = (\min(a,d)/\max(a,d))^{1/2}$. There is no scalar term for $m = P$, and (32) gives (37). □

Prime squares.

Proposition 3.4 (Prime-square simplification)**.** *Let $n \geq 8$ and let P be an odd prime with $P < N/64$. Then*

$$T_N(P^2) = 0, \qquad V_N(P^2) = \frac{1}{P} + O_k\big(N^{-1/2}\big). \tag{38}$$

Proof. We write $t = 2s$. Then the conditions become $|s| < P$ and $s^2 \equiv P^2 \pmod q$, with $q = N/16$. We know that both s and P are odd, so one of $s - P, s + P$ must have valuation exactly 1 and the other one must be divisible by $q/2 = N/32$. As a result, $s \equiv \pm P \pmod{N/32}$, and when $P < N/64$, this has no representative in $(-P, P)$. The endpoints are scalar terms. Hence $E_N(P^2) = 0$ by (35), and the result follows from (16), (15), and (32). □

Take an arbitrary odd P. Division by P reduces $s^2 \equiv P^2 \pmod q$ to the four square roots of 1 modulo $q \geq 8$. We can count these classes within $(-P, P)$ to get

$$T_N(P^2) \leq 4 \left(\frac{2P}{q} + 1 \right) = 128 \frac{P}{N} + 4, \tag{39}$$

so that (16) and (35) read

$$V_N(P^2) = \frac{c_N}{P} + E_N(P^2) + C_N(P^2), \qquad |E_N(P^2)| \leq K_\delta N^{-1/2} P^{2\delta} \left(128 \frac{P}{N} + 4 \right). \tag{40}$$

4. The unsplit one-level density

Write $h = \widehat{\phi}$. By (35), natural averaging requires counting pairs (P, t) as both indices grow with N.

Theorem 4.1. *Let ϕ be even Schwartz with Fourier support compactly contained in $(-3/2, 3/2)$ and $k \geq 2$ be an even integer. For all $n \geq 8$,*

$$U_N(\phi) = h(0) + \frac{\phi(0)}{2} + O_{k,\phi}\big(L^{-1}\big), \qquad N = 2^n, \quad L = \log(k^2 N). \tag{41}$$

More explicitly, select s, η such that $0 < s < 3/2$, $\operatorname{supp} h \subset [-s, s]$, and $0 < \eta < \min(1/2, 3/2 - s)$. Then for $n \geq 8$,

$$\left| U_N(\phi) - h(0) - \frac{\phi(0)}{2} \right| \leq C_{k,\phi,s,\eta} \left(L^{-1} + N^{s-3/2+\eta} + N^{-1/2+\eta} \right). \tag{42}$$

Counting primes and trace indices. Set $X = (k^2N)^s$ and $q = N/16$.

Lemma 4.2. *Let $n \geq 8$ and P be an odd prime. Then*

$$T_N(P) = \#\{r \in \mathbb{Z} : r^2 < P,\ q \mid P - r^2\}, \tag{43}$$

and $T_N(P) = 0$ for $P < q$. For all $B \geq q$,

$$\sum_{B/2<P\leq B} T_N(P) \leq 48\,\frac{B^{3/2}}{N}. \tag{44}$$

Proof. The congruence $N/4 \mid 4P - t^2$ forces $t = 2r$ and gives (43). When $P < q$, the positive difference $P - r^2$ cannot be a multiple of q. For $B \geq q$, there is a maximum of $2\sqrt{B} + 1 \leq 3\sqrt{B}$ possible r, and for each r at most B/q positive multiples $P - r^2$ of q. Counting all such integers P gives (44). □

The prime sums.

Lemma 4.3. *Let $n \geq 8$ and $0 < \delta < 1/2$. Then*

$$\frac{2}{L}\sum_{P\leq X} \frac{\log P}{\sqrt{P}}|E_N(P)|\left|h\left(\frac{\log P}{L}\right)\right| \leq \frac{96sK_\delta\sqrt{2}}{1-2^{-1-\delta}}\|h\|_\infty N^{-3/2}X^{1+\delta}, \tag{45}$$

and

$$\frac{2}{L}\sum_{P\leq X} \frac{\log P}{\sqrt{P}}|C_N(P)|\left|h\left(\frac{\log P}{L}\right)\right| \leq C_{k,\phi,s}N^{-1/2}. \tag{46}$$

Proof. On $B/2 < P \leq B$, equations (35) and (44), with $P^{-1/2} \leq \sqrt{2}B^{-1/2}$, give

$$\frac{2}{L}\sum_{B/2<P\leq B} \frac{\log P}{\sqrt{P}}|E_N(P)|\left|h\left(\frac{\log P}{L}\right)\right| \leq 96K_\delta\sqrt{2}\|h\|_\infty N^{-3/2}B^{1+\delta}\frac{\log B}{L}. \tag{47}$$

Using $\log B/L \leq s$, sum over $B = X, X/2, X/4, \ldots$ to get (45). For (46), equation (32) leaves the weight $(\log P)P^{-k/2}$. Its sum is bounded by $C_{\mathrm{Ch}}(1+\log X)$ for every $k \geq 2$ because of Chebyshev's bound $\sum_{P\leq y}\log P \leq C_{\mathrm{Ch}}y$ for $y \geq 2$ [MV06, Theorem 2.4] and partial summation. □

The square indices.

Lemma 4.4. *Let $n \geq 8$, $Z = X^{1/2}$, and $0 < \delta < 1/2$. Then*

$$\frac{2}{L}\sum_{P\leq Z} \frac{\log P}{P}|E_N(P^2)|\left|h\left(\frac{2\log P}{L}\right)\right| \leq \frac{C_\delta\|h\|_\infty}{L}\left(N^{-3/2}X^{1/2+\delta} + N^{-1/2}X^{\delta}\right), \tag{48}$$

and

$$\begin{aligned}\frac{2}{L}\sum_{P\leq Z} \frac{\log P}{P}|C_N(P^2)|\left|h\left(\frac{2\log P}{L}\right)\right| &\leq C_{k,\phi,s}N^{-1/2},\\ \frac{2}{L}\sum_{P\leq Z} \frac{\log P}{P}\frac{c_N}{P}\left|h\left(\frac{2\log P}{L}\right)\right| &\leq C_\phi L^{-1}.\end{aligned} \tag{49}$$

Proof. Once again, we use Chebyshev's bound and partial summation to get

$$\sum_{P\leq Z}(\log P)P^{2\delta} \leq C_{\mathrm{Ch}}Z^{1+2\delta}, \qquad \sum_{P\leq Z}(\log P)P^{-1+2\delta} \leq C_{\mathrm{Ch}}\left(1+\frac{1}{2\delta}\right)Z^{2\delta}, \tag{50}$$

and putting (50) into (40) yields (48) with

$$C_\delta = 256K_\delta C_{\mathrm{Ch}} + 8K_\delta C_{\mathrm{Ch}}\left(1+(2\delta)^{-1}\right).$$

For the scalar term use $c_N \le 2$ and $\sum_{P\ge 3}(\log P)P^{-2} < \infty$, and for the cusp term use (32) and the logarithmic bound from (46). □

Use Mertens' estimate [MV06, Theorem 2.7(b)] to evaluate the -1 in $\alpha_f(P)^2 + \beta_f(P)^2 = \lambda_f(P^2) - 1$.

$$\left| \sum_{\substack{P\le y \\ P \text{ odd}}} \frac{\log P}{P} - \log y \right| \le C_{\mathrm{M}} \qquad (y \ge 2), \tag{51}$$

where C_{M} is absolute. Partial summation gets

$$\left| \frac{2}{L} \sum_{P} \frac{\log P}{P} h\!\left(\frac{2\log P}{L}\right) - \int_0^\infty h(u)\,du \right| \le \frac{C'_{\mathrm{M}}(\|h\|_\infty + \|h'\|_1)}{L}, \tag{52}$$

including the initial interval $0 \le u \le 2\log 2/L$ in the error. Because h is even, $\int_0^\infty h(u)\,du = \phi(0)/2$.

A shared explicit formula.

Lemma 4.5. *Let ϕ be a Schwartz function on $\mathbb{R}$, and suppose that $h = \widehat{\phi}$ has support in $[-s, s]$ for some $s > 0$. Then ϕ extends uniquely to an entire function given by*

$$\phi(z) = \int_{-s}^{s} h(\xi) e^{2\pi i z\xi}\,d\xi. \tag{53}$$

For every integer $J \ge 0$ and every $z \in \mathbb{C}$,

$$|\phi(z)| \le C_{J,h}(1+|z|)^{-J} e^{2\pi s|\operatorname{Im} z|}, \qquad C_{J,h} = 2^J \max\left\{ \|h\|_1, (2\pi)^{-J}\|h^{(J)}\|_1 \right\}. \tag{54}$$

Proof. Because $h \in C_c^\infty(\mathbb{R})$, we can simply use Fourier inversion and differentiation under the integral to get the unique entire extension. We bound the integral directly for $|z| \le 1$ and integrate by parts J times for $|z| \ge 1$, and then all boundary terms are eliminated by compact support. □

Apply Shin–Templier [ST16, Proposition 4.4 and Equation 4.9] to f and $r \mapsto \phi(rL/(2\pi))$. The associated representation is cuspidal with trivial central character, conductor N, and archimedean parameters $-(k-1)/2$, $-(k+1)/2$ in their convention. Deligne gives their bound (4.4) with $\theta = 0$ and the completion $\Lambda(z, f) = N^{z/2}(2\pi)^{-z}\Gamma(z+(k-1)/2)L(z, f)$ has no poles. Lemma 4.5 supplies the Paley–Wiener test, and their zero-counting bound with (54) ensures absolute convergence over complex ordinates.

At an odd prime P the reciprocal Euler roots satisfy

$$\alpha_f(P) + \beta_f(P) = \lambda_f(P), \quad \alpha_f(P)\beta_f(P) = 1, \quad |\alpha_f(P)| = |\beta_f(P)| = 1. \tag{55}$$

At 2 the Euler factor is 1. Set $a_f(P^j) = \alpha_f(P)^j + \beta_f(P)^j$. Then, $a_f(P) = \lambda_f(P)$, $a_f(P^2) = \lambda_f(P^2) - 1$, and $|a_f(P^j)| \le 2$. In [ST16, (4.9)], expand on $\operatorname{Re} z = 2$ and shift the local integrals to $\operatorname{Re} z = 1/2$. The local factors are pole-free for $\operatorname{Re} z > 0$, and so is the gamma factor on this strip. Fourier inversion gets us

$$\begin{aligned} D(f;\phi) &= \mathcal{A}_{N,k}(\phi) - \frac{2}{L} \sum_{\substack{P \text{ odd} \\ P \text{ prime}}} \sum_{j\ge 1} \frac{\log P}{P^{j/2}} a_f(P^j) h\!\left(\frac{j\log P}{L}\right), \\ \mathcal{A}_{N,k}(\phi) &= \frac{\log(N/(4\pi^2))}{L} h(0) + \frac{2}{L}\int_{\mathbb{R}} \phi(x) \operatorname{Re}\frac{\Gamma'}{\Gamma}\!\left(\frac{k}{2} + \frac{2\pi i x}{L}\right) dx. \end{aligned} \tag{56}$$

Evenness produces the real part in the gamma term and complex-valued tests follow by linearity while compact support makes the prime-power sum finite.

The archimedean terms.

Lemma 4.6. *Let $a \geq 1$ be real and let $t \in \mathbb{R}$. Then*

$$\left|\frac{\Gamma'}{\Gamma}(a+it) - \frac{\Gamma'}{\Gamma}(a)\right| \leq 2|t|. \tag{57}$$

If a is an integer in addition, then $\frac{\Gamma'}{\Gamma}(a) = H_{a-1} - \gamma$, where $H_0 = 0$, $H_j = \sum_{r=1}^{j} r^{-1}$, and γ is Euler's constant.

Proof. By [DLM, (5.7.6)], everywhere except the nonpositive integers,

$$\frac{\Gamma'}{\Gamma}(z) = -\gamma + \sum_{r\geq 0}\left(\frac{1}{r+1} - \frac{1}{r+z}\right). \tag{58}$$

Subtracting the series at a termwise and bounding each difference by $|t|(r+a)^{-2}$ gives (57) from

$$\sum_{r\geq 0}(r+a)^{-2} \leq a^{-2} + \int_0^\infty (x+a)^{-2}\,dx = a^{-2} + a^{-1} \leq 2 \qquad (a \geq 1).$$

For a positive integer a the series (58) telescopes to $H_{a-1} - \gamma$. □

At $a = k/2 \geq 1$, Lemma 4.6 gives

$$|\mathcal{A}_{N,k}(\phi) - h(0)| \leq \frac{\log(4\pi^2 k^2)|h(0)| + 2|H_{k/2-1} - \gamma|\,\|\phi\|_1}{L} + \frac{8\pi}{L^2}\|x\phi(x)\|_1. \tag{59}$$

Geometric summation and integral comparison yield

$$\sum_{P \text{ odd}}\sum_{j\geq 3}\frac{\log P}{P^{j/2}} \leq \frac{1}{1-3^{-1/2}}\sum_{m\geq 3}\frac{\log m}{m^{3/2}} \leq \frac{6}{1-3^{-1/2}}. \tag{60}$$

Thus, natural averaging of (56) gives

$$\begin{aligned} U_N(\phi) = h(0) &- \frac{2}{L}\sum_{P \text{ odd}}\frac{\log P}{\sqrt{P}}V_N(P)h\left(\frac{\log P}{L}\right) \\ &- \frac{2}{L}\sum_{P \text{ odd}}\frac{\log P}{P}\big(V_N(P^2)-1\big)h\left(\frac{2\log P}{L}\right) + O_{k,\phi}(L^{-1}). \end{aligned} \tag{61}$$

The support of h restricts the sums to $P \leq X$ and $P \leq Z = X^{1/2}$, respectively.

Proof of Theorem 4.1. The -1 in the square sum in (61) contributes $\phi(0)/2 + O_\phi(L^{-1})$ by (52). Lemmas 4.3 and 4.4 bound the remaining terms, so for $0 < \delta < 1/2$,

$$\begin{aligned} \left|U_N(\phi) - h(0) - \frac{\phi(0)}{2}\right| \leq C_{k,\phi,s,\delta}\Big[&L^{-1} + N^{-1/2} + N^{-3/2}(k^2N)^{s(1+\delta)} \\ &+ L^{-1}N^{-3/2}(k^2N)^{s(1/2+\delta)} + L^{-1}N^{-1/2}(k^2N)^{s\delta}\Big]. \end{aligned} \tag{62}$$

Given s and η as in the statement, choose

$$\delta = \frac{\eta}{2(1+s)}. \tag{63}$$

It satisfies $0 < \delta < 1/2$ and $s\delta < \eta/2$, and the exponents in (62) also satisfy

$$\begin{aligned} -\frac{3}{2} + s(1+\delta) &\le s - \frac{3}{2} + \eta, \\ -\frac{3}{2} + s\left(\frac{1}{2}+\delta\right) &\le -\frac{1}{2} + \eta, \\ -\frac{1}{2} + s\delta &\le -\frac{1}{2} + \eta. \end{aligned} \tag{64}$$

Hence (42) holds, and the negative exponents of N imply (41). □

5. The signed one-level density

Write $H = H_k^{\mathrm{new}}(N)$, $H^{\pm} = \{f \in H : \varepsilon_f = \pm 1\}$, and $A_P = \sum_{f\in H} \varepsilon_f \lambda_f(P)\sqrt{P}$ for odd primes P. In discriminants $D = df^2$, d is the fundamental part and f is the square factor.

The trace and sign normalisation. Set $\varphi_{t,vm} = \arcsin(t/(2\sqrt{vm}))$. The roots of $x^2 - (t/\sqrt{v})x + m$ are $e^{\pm i(\pi/2-\varphi_{t,vm})}\sqrt{m}$. Hence (1) gives

$$p_{k-2}(t/\sqrt{v}, m) = \frac{2(-1)^{k/2+1} m^{(k-1)/2}\sqrt{v}\cos\big((k-1)\arcsin(t/(2\sqrt{vm}))\big)}{\sqrt{4vm - t^2}} \qquad (t^2 < 4vm), \tag{65}$$

as in [BKL+26, (2.5)]. The signed trace formula of Skoruppa–Zagier [SZ88, p. 117 and §1, Theorem 1], in the form [BKL+26, (2.2)–(2.4)], states that, for $N \ge 1$, even $k \ge 2$ and $(m, N) = 1$,

$$\begin{aligned} \operatorname{tr}(T_m W_N, S_k^{\mathrm{new}}(N)) &= \sum_{M\mid N} \alpha(N/M)\mathfrak{s}_{k,M}(m), \\ \mathfrak{s}_{k,M}(m) &= -\frac{1}{2}\sum_{v\mid M} \sum_{\substack{t\in\mathbb{Z},\ v\mid t,\ t^2\le 4vm \\ ((t/v)^2, M/v)\ \text{squarefree}}} p_{k-2}(t/\sqrt{v}, m) H(t^2 - 4vm) \\ &\quad - \frac{1}{2}\sum_{d\mid m} \min(d, m/d)^{k-1} \gcd(Q(M), d + m/d) + \mathbf{1}_{k=2}\tau(M)\sigma_1(m), \\ \alpha(p^j) &= \begin{cases} 1, & j = 0, 3, \\ -1, & j = 1, 2, \\ 0, & j \ge 4, \end{cases} \qquad Q(M) = \prod_{p\mid M} p^{\lfloor v_p(M)/2\rfloor}, \\ H(0) &= -\frac{1}{12}, \qquad H(D) = \frac{\sqrt{|D|}}{\pi} L(1, \psi_D) \quad (D < 0). \end{aligned} \tag{66}$$

At a polynomial root with multiplicity > 1, $p_{k-2}(x, m) = (k-1)(x/2)^{k-2}$. The weight parameter in [BKL+26] is $k/2 + 1$ and its exceptional value 2 corresponds to $k = 2$ here.

The eigenvalues of T_m and W_{2^a} give, for every odd m and $H = H_k^{\mathrm{new}}(2^a)$,

$$\sum_{f\in H} \varepsilon_f \lambda_f(m) = (-1)^{k/2} m^{(1-k)/2} \operatorname{tr}\big(T_m \circ W_{2^a}, S_k^{\mathrm{new}}(2^a)\big). \tag{67}$$

Write $R_N(m) = \sum_{f\in H} \varepsilon_f \lambda_f(m)$ and $R_N = R_N(1)$. In (66), set $M = N/2^j$, $0 \le j \le 3$, and $\alpha_j = (1, -1, -1, 1)_j$. Hence, $M \ge 4$ for $n \ge 5$. At index 1, the nonzero admissible pairs (v, t) are $(1, \pm 1), (2, \pm 2), (4, \pm 4)$. The conditions $v \mid t$ and $t^2 \le 4v$ force $v \le 4$. $(1, \pm 2)$ is excluded because the gcd is divisible by 4 and is consequently not squarefree. Each listed contribution has the same value at all four levels, so the sieve cancels it. This includes the zero discriminant pair

$(4, \pm 4)$, where the polynomial equals $k-1$. At $t=0$ only $v = M, M/2$ survive. The hyperbolic term is -1 and the weight-two correction is linear in j, so

$$\begin{aligned} \sum_{j=0}^{3} \alpha_j = \sum_{j=0}^{3} j\alpha_j = 0, \qquad p_{k-2}(0,1) = (-1)^{k/2-1}, \\ R_N = \tfrac{1}{2}\big\{H(-4N) - 2H(-N) + H(-N/4)\big\}. \end{aligned} \tag{68}$$

Over one period of each character, a geometric series and integration of the sums give

$$\begin{aligned} L(1, \psi_{d2^{2e}}) &= (2 - 2^{-e}) L(1, \chi_d), \\ L(1, \chi_{-4}) &= \int_0^1 \frac{dx}{1+x^2} = \frac{\pi}{4}, \\ L(1, \chi_{-8}) &= \int_0^1 \frac{1+x^2}{1+x^4}\, dx = \frac{\pi}{2\sqrt{2}}. \end{aligned} \tag{69}$$

The first identity holds for $d = -4, -8$. The integrals are justified by dominated convergence, and the last has antiderivative $2^{-1/2} \arctan(x\sqrt{2}/(1-x^2))$. The class-number formula yields

$$\begin{aligned} H(-2^{2e+2}) = 2^e - \tfrac{1}{2}, \qquad H(-2^{2e+3}) = 2^{e+1} - 1, \\ R_{2^{2a}} = 2^{a-3} \quad (a \geq 3), \qquad R_{2^{2a+1}} = 2^{a-2} \quad (a \geq 2). \end{aligned} \tag{70}$$

Combining the dimension formula in Lemma 2.5 with (70) gives

$$\begin{aligned} |H| \geq \frac{k-1}{64} N, \qquad 0 \leq \frac{R_N}{|H|} \leq \frac{8}{k-1} N^{-1/2}, \\ \left| \frac{|H|}{2|H^{\pm}|} - 1 \right| \leq \frac{16}{k-1} N^{-1/2} \qquad (n \geq 7). \end{aligned} \tag{71}$$

Both sign classes are nonempty, because here $R_N/|H| \leq 1/2$ for $n \geq 7$. At $(n,k) = (5,2), (6,2)$, $|H| = R_N = 1$ and D_N^- is undefined.

For odd m, write $t = vu$. The gcd restriction allows both even and odd when $M/v = 1, 2$, but only odd u when $M/v \geq 4$. Summing α_j leaves

v	u allowed	coefficient
N	all	1
$N/4$	u odd	-1
$N/4$	u even	-2
$N/16$	u even	1.

(72)

With $b = \log_2(N/v)$, a stratum occurs exactly for $j \leq b$, and when u is even, it is subject also to $b - j \leq 1$.

For odd p, at $m = p^2$, the hyperbolic and weight-two terms cancel because the second arguments $p^2+1, 2p$ of the gcds in the hyperbolic term have 2-adic valuation 1. A zero discriminant requires $v = 1$ or 4. The former is inadmissible, and the latter cancels as above, with $H(0) = -1/12$ retained.

Lemma 5.1. *Let p be an odd prime and $\delta > 0$. Then*

$$\begin{aligned} |p_{k-2}(x,m)| \leq (k-1) m^{(k-2)/2} \qquad (|x| \leq 2\sqrt{m}), \\ |R_N(p^2)| \leq K_{k,\delta} \big\{ N^{1/2+\delta} p^{2\delta} + N^{\delta} p^{1+2\delta} \big\}, \\ K_{k,\delta} = 48(k-1) C_{\delta}^{\mathrm{cl}} 4^{1/2+\delta}. \end{aligned} \tag{73}$$

Proof. The polynomial is a sum of $k-1$ products of roots of modulus $\sqrt{m}$. Each stratum in (72) contains at most $1+16\sqrt{m/N}$ indices and has $|D| \le 4Nm$. Applying Lemma 2.2 with constant C_δ^{cl} gives the second bound. □

The dyadic trace specializations. For $q = 2^b$, summing over the odd residue classes gives

$$c_q(t) = q\mathbf{1}_{q|t} - \frac{q}{2}\mathbf{1}_{q/2|t}, \qquad c_q((q/2)r) = \frac{q}{2}(-1)^r. \tag{74}$$

Scaling class numbers at 2 combines the four strata in (72) into the following prime traces.

Proposition 5.2. *Let P be odd and prime. At even depth $n \ge 8$, set $q = 2^{n/2}$. Then*

$$\begin{aligned} A_P = \frac{2\sqrt{P}}{\pi} w(P) \sum_{t^2<4P} c_q(t)\cos((k-1)\varphi_{t,P}) L(1,\psi_{t^2-4P}) \\ + \frac{(-1)^{k/2+1}}{2} c_q(P+1)P^{1-k/2}, \end{aligned} \tag{75}$$

$P \pmod 8$	1	3	5	7
$w(P)$	1/2	3/8	1/2	1/4

At odd depth $n \ge 7$, let $Q = 2^{(n+1)/2}$. Then

$$A_P = \frac{\sqrt{P}}{\pi\sqrt{2}} \sum_{t^2<8P} c_Q(t)\cos((k-1)\varphi_{t,2P}) L(1,\psi_{t^2-8P}). \tag{76}$$

Proof. For a negative discriminant $D = d(2^e f_0)^2$ with f_0 odd, set $\alpha = (d/2)$ and $T_e = 1 + (2-\alpha)(2^e-1)$. The prime-power definition of ψ_D and a finite geometric sum give

$$\begin{aligned} H(D) &= T_e H(df_0^2), \\ H(4^jD) - H(4^{j-1}D) &= 2^{j-1}\{H(4D) - H(D)\} \quad (j \ge 1), \\ H(4D) - H(D) &= \frac{(2-\alpha)2^e}{T_e} H(D). \end{aligned} \tag{77}$$

The local class-number ratio is $2^e[1 + (1-\alpha)\sum_{b=1}^{e} 2^{-b}] = T_e$. $D < 0$ ensures a finite primitive L-value. Since v is a power of 2 and $vu^2 = 4P$ has no integer solution, at a prime index the discriminant is nonzero. For even n, substituting $t = u\sqrt{v}$ in each stratum makes the polynomial $p_{k-2}(t,P)$ and the discriminant $v(t^2-4P)$. With $D_t = t^2 - 4P$, the resulting elliptic trace is

$$\begin{aligned} \mathrm{tr}(T_P W_N)^{\mathrm{ell}} &= \frac{1}{2} \sum_{t^2<4P} p_{k-2}(t,P)\mathcal{H}_{n,t}, \\ \mathcal{H}_{n,t} &= \mathbf{1}_{q/2|t}\{H(2^{n-2}D_t) - H(2^{n-4}D_t)\} - \mathbf{1}_{q|t}\{H(2^nD_t) - H(2^{n-2}D_t)\} \\ &= -\frac{1}{2} c_q(t)\{H(4D_t) - H(D_t)\}. \end{aligned} \tag{78}$$

Group the strata as all u at N, minus all u at $N/4$, minus even u at $N/4$, plus even u at $N/16$. Then apply (77) at $j = n/2-1, n/2$. The hyperbolic trace is

$$-\gcd(q, P+1) + 2\gcd(q/2, P+1) - \gcd(q/4, P+1) = -\frac{1}{2} c_q(P+1). \tag{79}$$

This follows in the three cases $v_2(P+1) < n/2-1$, $= n/2-1$, and $\ge n/2$, and the weight-two term cancels by (68).

Equations (65) and (67) along with the class-number formula give (75) with $w(t,P) = (2-\alpha)2^e/(4T_e)$. On the support $q/2 \mid t$, hence $8 \mid t$. Write $D_t = 4K$. Then, we have $K = (t/2)^2 - P \equiv -P \pmod 8$. If $P \equiv 1 \pmod 4$, the fundamental discriminant has valuation

2, so $(e,\alpha)=(0,0)$ and $w=1/2$. Otherwise, if $P\equiv 3 \pmod 4$, the fundamental discriminant is odd, $e=1$ and $\alpha=(K/2)$. The two cases $K\equiv 5,1 \pmod 8$ give $w=3/8,1/4$, respectively. For odd n, use $t=u\sqrt{2v}$ instead to make the polynomial $p_{k-2}(t/\sqrt{2},P)$ and the discriminant $(v/2)(t^2-8P)$. Setting $D_t=t^2-8P$ gives

$$\begin{aligned}\operatorname{tr}(T_P W_N)^{\mathrm{ell}} &= \frac{1}{2}\sum_{t^2<8P} p_{k-2}(t/\sqrt{2},P)\mathcal{J}_{n,t},\\ \mathcal{J}_{n,t} &= \mathbf{1}_{Q/2|t}\{H(2^{n-3}D_t)-H(2^{n-5}D_t)\}-\mathbf{1}_{Q|t}\{H(2^{n-1}D_t)-H(2^{n-3}D_t)\}\\ &= -\frac{1}{4}c_Q(t)\{H(4D_t)-H(D_t)\}=-\frac{1}{2}c_Q(t)H(D_t).\end{aligned} \tag{80}$$

Use the same grouping and (77) at $j=(n-3)/2,(n-1)/2$. Since $8\mid t$, $v_2(D_t)=3$ and $(e,\alpha)=(0,0)$, which gives us the last equality. The hyperbolic gcds cancel in pairs and the weight-two terms cancel as before. Equations (65) (with level divisor 2) and (67) give (76). □

Finite coefficient averages and their tails. To average the trace weight, define for $\chi\in\{1,\chi_{-4},\chi_8,\chi_{-8}\}$,

$$q_m=\operatorname{lcm}(8,m^2),\qquad \widetilde{\psi}_{t,\chi}(m)=\frac{1}{\varphi(q_m)}\sum_{\substack{x \bmod q_m\\(x,2m)=1}}\chi(-x)\psi_{t^2-4x}(m),\qquad L(s,\widetilde{\psi}_{t,\chi})=\sum_{m\ge 1}\frac{\widetilde{\psi}_{t,\chi}(m)}{m^s}. \tag{81}$$

For positive nonsquare D, use the same factorization $D=df^2$ with d fundamental and the same Kronecker definition of ψ_D. For nonzero squares take $d=1$, and set $\psi_0(m)=1$. At an odd prime p, the value of $\psi_D(p^b)$ is 1 if $p^{2b}\mid D$, 0 if $v_p(D)<2b$ is odd, and $(D/p^{2e}\mid p)^{b-e}$ when $v_p(D)=2e<2b$. Thus it depends only on $D \bmod p^{2b}$. At 2, for $D=4K$ with $K=2^j u$ and u odd, the conditions determining that value are

condition	$v_2(f)$ in $D=df^2$	$(d/2)$
$j=2e, u\equiv 1 \pmod 4$	$e+1$	$(u/2)$
$j=2e, u\equiv 3 \pmod 4$	e	0
$j=2e+1$	e	0.

(82)

Thus $\psi_D(2^b)=(d/2)^{\max(b-v_2(f),0)}$ is determined by $K \bmod 2^{2b}$. If these residues do not determine the symbol, its exponent is zero. For odd $D=t^2-4x$, the symbol is determined modulo 8. This proves independence of the choice of residue-class representatives in (81), including both zero and square discriminants.

Lemma 5.3. *For every $t\in\mathbb{Z}$ the odd local factors are*

$$E_p(z;t)=\begin{cases}(1-z^2)^{-1}, & p\mid t,\\ \dfrac{p^2-(p+1)z-(p+1)z^2}{(1-z^2)(p^2-z)}, & p\nmid t,\end{cases}\qquad (p>2). \tag{83}$$

For $\operatorname{Re} s>1/2$ the following series and products converge absolutely:

$$\begin{aligned}L(s,\widetilde{\psi}_{t,\chi}) &= E_{2,\chi}(2^{-s};t)\mathcal{G}_t(s),\\ E_{2,\chi}(z;t) &= \sum_{b\ge 0}\widetilde{\psi}_{t,\chi}(2^b)z^b,\qquad \mathcal{G}_t(s)=\prod_{p>2}E_p(p^{-s};t),\\ \mathcal{G}_t &:= \mathcal{G}_t(1)=\frac{\pi^2}{8}\prod_{p\nmid 2t}\frac{p^2-p-1}{p(p-1)}.\end{aligned} \tag{84}$$

In addition, uniformly in t, χ and $x \geq 1$,

$$\sum_{m\geq 1} \frac{|\widetilde{\psi}_{t,\chi}(m)|}{m}(1-e^{-m/x}) \leq C^{\mathrm{tail}} x^{-1/2},$$

$$C^{\mathrm{tail}} = \frac{4}{1-2^{-1/2}} \prod_{p>2} \left(1 + \frac{1}{\sqrt{p}(p-1)}\right) < \infty, \tag{85}$$

$$\sum_{m\geq 1} \frac{|\widetilde{\psi}_{t,\chi}(m)|}{m} \leq \frac{C^{\mathrm{tail}}}{1-e^{-1}}.$$

Proof. By the Chinese remainder theorem, we factor the average into local averages. The local average at 2 has absolute value at most 1, but isn't necessarily equal to 1 at $b = 0$. Thus, the twisted sequence need not be multiplicative.

If $p \mid t$, the Legendre symbol of $t^2 - 4x$ has mean zero over units, giving local average 1 in even degrees and 0 in odd degrees. If $p \nmid t$, let U_b average $\psi_D(p^b)$ over all $D \bmod p^{2b}$, with $U_0 = 1$. Splitting according to $v_p(D) = 0, 1$ or ≥ 2 gives

$$U_b = (1-p^{-1})\mathbf{1}_{2|b} + p^{-2}U_{b-1}, \qquad 0 \leq U_b \leq 1,$$

$$a_p(b;t) = \begin{cases} \dfrac{p-2}{p-1} + \dfrac{U_{b-1}}{p(p-1)}, & b \geq 1 \text{ even}, \\ -\dfrac{1}{p-1} + \dfrac{U_{b-1}}{p(p-1)}, & b \text{ odd}. \end{cases} \tag{86}$$

To obtain the second line, exclude $D \equiv t^2 \pmod p$, whose symbol is 1, and divide by $1-p^{-1}$. Summing gives $\sum_{b\geq 0} U_b z^b = (1-z^2/p)/((1-z^2)(1-z/p^2))$, hence (83) and the bounds $|a_p(2j;t)| \leq 1$, $|a_p(2j+1;t)| \leq 1/(p-1)$ for $j \geq 0$.

Write $m = 2^b uv^2$ uniquely with u and v odd and u squarefree. The preceding bounds give

$$|\widetilde{\psi}_{t,\chi}(2^b uv^2)| \leq \frac{1}{\varphi(u)}, \qquad \sigma > 1/2,$$

$$\sum_{m\geq 1} \frac{|\widetilde{\psi}_{t,\chi}(m)|}{m^\sigma} \leq \frac{\zeta(2\sigma)}{1-2^{-\sigma}} \prod_{p>2} \left(1 + \frac{1}{p^\sigma(p-1)}\right). \tag{87}$$

At $s = 1$, each odd factor is $(1-p^{-2})^{-1}$ if $p \mid t$ or $(1-p^{-2})^{-1}(p^2-p-1)/(p(p-1))$ if $p \nmid t$. The identity $\prod_{p>2}(1-p^{-2})^{-1} = \pi^2/8$ gives (84), including at $t = 0$.

For $T > 0$, splitting at $v = \sqrt{T}$ and comparing the tail with an integral gives

$$\sum_{v\geq 1} v^{-2} \min(1, v^2/T) \leq 4T^{-1/2}. \tag{88}$$

For $T < 1$ use $\zeta(2) < 2$, otherwise the two parts are bounded by $T^{-1/2}$ and $2T^{-1/2}$. With $T = x/(2^b u)$ and $1 - e^{-y} \leq \min(1, y)$, summation over b and u gives the stated tail constant. Its product converges by comparison with $\sum_{j\geq 3} j^{-3/2}$. The final bound follows at $x = 1$ from $1 - e^{-m} \geq 1 - e^{-1}$. □

Lemma 5.4. *If $8 \mid t$, then*

$$E_{2,1}(z;t) = \frac{2+z-2z^2}{2(1-z^2)}, \qquad E_{2,\chi_{-4}}(z;t) = \frac{z}{2(1-z^2)},$$

$$E_{2,\chi_8}(z;t) = E_{2,\chi_{-8}}(z;t) = \frac{z^2}{2(1-z^2)}. \tag{89}$$

The weight in (75) *satisfies*

$$w(P) = \frac{13 - 3\chi_{-4}(-P) - \chi_8(-P) - \chi_{-8}(-P)}{32},$$
$$\frac{13}{32}L(1,\widetilde{\psi}_{t,1}) - \frac{3}{32}L(1,\widetilde{\psi}_{t,\chi_{-4}}) - \frac{1}{32}L(1,\widetilde{\psi}_{t,\chi_8}) - \frac{1}{32}L(1,\widetilde{\psi}_{t,\chi_{-8}}) = \frac{1}{2}\mathcal{G}_t. \tag{90}$$

Proof. Write $t^2 - 4x = 4K$, with $K \equiv -x \pmod 8$, and K odd. By (82), for $b \geq 1$ the four classes $K \equiv 1, 5, 3, 7$ contribute $1, (-1)^{b-1}, 0,$ and 0, respectively, each with frequency $1/4$. Thus

$$\widetilde{\psi}_{t,\chi}(1) = \mathbf{1}_{\chi=1}, \qquad \widetilde{\psi}_{t,\chi}(2^b) = \frac{1 + \chi(5)(-1)^{b-1}}{4} \quad (b \geq 1). \tag{91}$$

The values $\chi(5)$ are $1, 1, -1, -1$, so geometric summation gives (89). The four odd residue classes give the weight expansion. At $z = 1/2$, the factors are $4/3, 1/3, 1/6, 1/6$, whose weighted sum is $13/24 - 1/32 - 1/192 - 1/192 = 1/2$. Equation (84) completes the proof. □

Uniform prime averaging. Assume the Riemann hypothesis for every primitive Dirichlet L-function, including ζ. Fix $t \in \mathbb{Z}$, $\chi \in \{1, \chi_{-4}, \chi_8, \chi_{-8}\}$, and $t^2/4 \leq A < B$ for $B \geq 3$. Let Φ be continuous on $[A, B]$ and C^1 on $(A, B]$, with $M = \max|\Phi|$ and $V = \int_A^B |\Phi'| < \infty$.

Applying [IK04, Theorem 5.15] to the primitive inducing character gives

$$\Big|\sum_{p \leq u} \nu(p)\log p - \delta_\nu u\Big| \leq C_{\mathrm{abs}} u^{1/2}\log^2(2qu) \qquad (u \geq 2) \tag{92}$$

for every character ν modulo q, with δ_ν the indicator of the principal character. The inducing primitive L-function has the required degree-one analytic properties, local parameters of modulus at most 1, and the assumed RH. Removing prime powers contributes an error of at most $C_{\mathrm{abs}}\sqrt{u}\log^2(2u)$ and removing primes dividing q contributes an error of at most $C_{\mathrm{abs}}\log(2q)\log(2u)$, both within the displayed bound.

We bound the character expansion by the ℓ^1 norm of its coefficients. For a function g on the finite abelian group $G_Q = (\mathbb{Z}/Q\mathbb{Z})^\times$, character orthogonality [Apo76, Theorems 6.11–6.12] and Cauchy–Schwarz give

$$\widehat{g}(\nu) = \frac{1}{|G_Q|}\sum_{x \in G_Q} g(x)\overline{\nu(x)}, \qquad \beta(g) = \sum_\nu |\widehat{g}(\nu)|, \qquad \beta(g) \leq \sqrt{\#\operatorname{supp} g} \quad (|g| \leq 1). \tag{93}$$

Inflation and multiplication by a character preserve β. For products over the factors in the Chinese remainder decomposition, the norms multiply, and coset indicators have norm 1.

For an odd prime p set $D = t^2 - 4x$ with x a unit. The definition of ψ_D, with the stated conventions at squares and at zero, gives

$$\psi_D(p^j) = \mathbf{1}_{p^{2j}\mid D} + \sum_{e=0}^{j-1} \mathbf{1}_{v_p(D)=2e}\left(\frac{D/p^{2e}}{p}\right)^{j-e}. \tag{94}$$

The summand indexed by e depends only on the residue class modulo p^{2e+1} and is nonzero on at most $p-1$ such classes, while the first term is either zero or a coset indicator. Hence (93) proves

$$\beta\big(\psi_{t^2-4x}(p^j)\big) \leq 1 + j\sqrt{p} \leq (j+1)p^{1/2}. \tag{95}$$

If $p \mid t$ then D is a unit and the function is the character $(-4x/p)^j$, of norm 1. Replacing 4 with 8 leaves this entire argument valid at odd primes.

At 2, if t is odd, then $D \equiv 5 \pmod 8$ and the symbol is constant. If $4 \mid t$, the function depends only on x mod 8 and has norm at most 2. For $t = 2u$, where u is odd, set $z = u^2 - x$. The possible values $0, 2, 3$ of the 2-adic valuation of a fundamental discriminant give

$$\psi_{4z}(2^j) = \mathbf{1}_{2^{2j}\mid z} + \sum_{a=0}^{j-1} \mathbf{1}_{v_2(z)=2a}\, \mathbf{1}_{z/2^{2a}\equiv 1 \ (4)} \left(\frac{z/2^{2a}}{2}\right)^{\max(j-a-1,0)}. \tag{96}$$

If the odd part of $z/2^{2a}$ is 1 modulo 4, the conductor exponent is $a+1$. Otherwise, and at odd valuations of z, the fundamental character vanishes at 2 once the conductor exponent is passed. Higher valuations give the first indicator. Since z is even, the stratum $a = 0$ is empty. The other strata require modulus 2^{2j} at $a = j-1$ and 2^{2a+3} at $a \le j-2$, with at most eight residue classes on which each stratum is nonzero. Consequently

$$\beta\big(\psi_{t^2-4x}(2^j)\big) \le 1 + j\sqrt{8} \le 4(j+1), \qquad \beta_m := \beta\big(\chi(-x)\psi_{t^2-4x}(m)\big) \le 4\tau(m)m^{1/2} \le C_\delta m^{1/2+\delta}. \tag{97}$$

For the last bound, use $j+1 \le 2^j \le p^{\delta j}$ when $p^\delta \ge 2$. Then, the remaining primes contribute the finite product $\prod_{p^\delta<2} \sup_{j\ge 0}(j+1)p^{-\delta j}$. The functions depend only on the residue class modulo $q_m = \operatorname{lcm}(8, m^2)$, uniformly in t.

The principal character coefficient is $\widetilde{\psi}_{t,\chi}(m)$. The endpoint hypotheses imply absolute continuity by applying the fundamental theorem of calculus on $[A+h, B]$ and letting $h \downarrow 0$. Character expansion and partial summation in (92) give

$$\begin{aligned}\Big|\sum_{A<P\le B,\ (P,2m)=1} \Phi(P)\chi(-P)\psi_{t^2-4P}(m)\log P - \widetilde{\psi}_{t,\chi}(m)\int_A^B \Phi\Big| \\ \le C_{\mathrm{abs}}(M+V)\beta_m B^{1/2}\log^2(2Bm).\end{aligned} \tag{98}$$

After exponential smoothing, restoring the omitted primes costs

$$M\sum_{m\ge 1}\frac{e^{-m/x}}{m}\sum_{P\mid 2m}\log P \le C_{\mathrm{abs}} M\log^2(2x) \qquad (x \ge 1), \tag{99}$$

since the inner sum is at most $\log(2m)$ and integral comparison bounds the remaining series.

For a primitive nonprincipal character ν of conductor q, [IK04, Corollary 5.20] gives

$$|L(\sigma+iv,\nu)| \le C_\eta\{q(2+|v|)\}^\eta \qquad (1/2 \le \sigma \le 2). \tag{100}$$

The required RH, Ramanujan bound and degree-one analytic properties hold. For $D = df^2 < 0$, χ_d is nonprincipal and the prime-power coefficients give

$$L(s,\psi_{df^2}) = L(s,\chi_d)\prod_{p^e\,\|\,f}\left[1 + (1-\chi_d(p))\sum_{j=1}^{e} p^{-js}\right], \qquad \Big|\prod_{p^e\,\|\,f}[\cdots]\Big| \le C_\eta f^\eta \quad (\operatorname{Re} s \ge 1/2). \tag{101}$$

Each factor is at most $1 + 2/(\sqrt{p}-1) \le p^\eta$ for sufficiently large p, and the remaining factors are bounded independently of e.

Apply Mellin inversion [DLM, (2.5.2)] to e^{-u} and move the integral of $\Gamma(w)L(1+w,\psi_D)x^w$ from $\operatorname{Re} w = 1$ to $-1/2+\eta$, $0 < \eta < 1/4$. Only the pole of Γ at 0 is crossed. Stirling's formula [DLM, (5.11.3), (5.11.9)] gives exponential decay on the vertical lines and horizontal segments. (100)–(101), with reduced exponents, then give

$$\big|L(1,\psi_D) - S_x(D)\big| \le C_\eta |D|^\eta x^{-1/2+\eta} \qquad (D<0,\ x\ge 1), \tag{102}$$

where $S_x(D) = \sum_m \psi_D(m)e^{-m/x}/m$.

For each counted prime, $A \ge t^2/4$ gives $D < 0$ and $|D| = 4P - t^2 \le 4B$. For $1 \le x \le B^{1/2}$, summing (98) with weight $e^{-m/x}/m$, using (97) and the absolute coefficient tail (85) gives

$$|E| \le C_\eta \left\{ MB^{1+\eta}x^{-1/2} + (M+V)B^{1/2+\eta}x^{1/2+\eta} \right\}. \tag{103}$$

For the first term, (102) and Chebyshev give $C_\eta MB^{1+2\eta}x^{-1/2+\eta}$ and we use $x^\eta \le B^{\eta/2}$ and replace η by $2\eta/5$. The coefficient tail contributes $C^{\mathrm{tail}}MBx^{-1/2}$, and the logarithms in (98)–(99) are absorbed into B^η.

Theorem 5.5. *Assume the Riemann hypothesis for every primitive Dirichlet L-function. Let χ be one of the four characters $1, \chi_{-4}, \chi_8, \chi_{-8}$, $t \in \mathbb{Z}$, $B \ge 3$, $A \ge t^2/4$, and Φ be continuous on $[A,B]$ and C^1 on $(A,B]$ with $M = \max|\Phi|$ and $V = \int_A^B |\Phi'| < \infty$. Then, for every $\epsilon > 0$,*

$$\Big| \sum_{A<P\le B} \Phi(P)\chi(-P)L(1,\psi_{t^2-4P}) \log P - L(1,\widetilde{\psi}_{t,\chi}) \int_A^B \Phi(u)\,du \Big| \le C_\epsilon M^{1/2}(M+V)^{1/2}B^{3/4+\epsilon}, \tag{104}$$

uniformly in t, χ, A and Φ.

Proof. If $M = 0$ the assertion is immediate. Otherwise set $R = (M+V)/M$. If $R \le B^{1/2}$, choose $x = B^{1/2}/R$ in (103) and take $\eta \le \epsilon/3$. If $R > B^{1/2}$, Lemma 2.2 combined with Chebyshev's bound gives $C_\eta MB^{1+\eta}$ for the whole discrepancy, because $|D| \le 4B$ and the main coefficient is uniformly bounded. In that case $M^{1/2}(M+V)^{1/2}B^{3/4} \ge MB$. □

The prime sum at odd depth. For the odd-depth discriminant $t^2 - 8P$, valuation 3 at 2 removes the even coefficients and leaves the following odd-modulus average.

Theorem 5.6. *Assume the Riemann hypothesis for every primitive Dirichlet L-function. Let $8 \mid t$, $B \ge 3$, and $A \ge 2, t^2/8$. Let Φ be continuous on $[A,B]$, C^1 on $(A,B]$, with $M = \max|\Phi|$ and $V = \int_A^B |\Phi'| < \infty$. Then, for every $\epsilon > 0$,*

$$\Big| \sum_{A<P\le B} \Phi(P)L(1,\psi_{t^2-8P}) \log P - \mathcal{G}_t \int_A^B \Phi \Big| \le C_\epsilon M^{1/2}(M+V)^{1/2}B^{3/4+\epsilon}. \tag{105}$$

Proof. The hypothesis $A \ge 2$ restricts the sum to odd primes. For every counted odd prime, $v_2(t^2-8P) = 3$, the square factor f is odd, and $\psi_D(m) = 0$ at even m. For odd m, first project the average over invertible residue classes to the units modulo m^2. The coefficient is independent of the coordinate modulo 8. On that odd modulus, multiplication by 2 is a bijection and identifies the mean with the untwisted odd coefficient in (81). Evaluating the Dirichlet series of these averaged coefficients at $s = 1$ therefore gives $\mathcal{G}_t$, by (84), with 1 being the local factor at 2. Its absolute tail is bounded by restricting (85) for $\chi = 1$ to odd m. The Fourier argument (94)–(99) is unchanged at odd primes. Apply (102) with $|D| \le 8B$ and Chebyshev's bound, and repeat the optimization of Theorem 5.5. □

Uniform summation over the trace indices. Define, for any prime ℓ, $s_k = (-1)^{k/2+1}$ and $U_j(\cos\vartheta)\sin\vartheta = \sin((j+1)\vartheta)$:

$$\begin{aligned} F_k(u) &= U_{k-2}(u)\sqrt{1-u^2} \quad (|u|<1), \qquad F_k(u) = 0 \quad (|u| \ge 1), \\ \mathcal{M}_\ell(y) &= \frac{s_k 2\pi\sqrt{y}}{(k-1)(1-1/\ell)} \sum_{r\in\mathbb{Z}} (\mathbf{1}_{\ell\mid r} - 1/\ell)\mathfrak{p}_\ell(r) F_k\left(\frac{r}{2\ell\sqrt{y}}\right), \\ \mathfrak{p}_\ell(r) &= \prod_{p\nmid \ell r}\left(1 - \frac{1}{p(p-1)}\right), \qquad \mathfrak{p}_\ell(0) = 1. \end{aligned} \tag{106}$$

Write $M_{k,2} = \mathcal{M}_2$.

Proposition 5.7 (Bound for the integrated density error)**.** *For every fixed even integer $k \geq 2$ and every prime ℓ, the function in (106) satisfies*

$$\begin{aligned} |\mathcal{M}_\ell(v)| &\leq C_{k,\ell} v^{1/2} \qquad (0 < v \leq 1), \\ P_\ell(V) := \int_0^V \left\{\mathcal{M}_\ell(v) - \tfrac{1}{2}\mathbf{1}_{v\geq 1}\right\} \frac{dv}{v}, &\qquad |P_\ell(V)| \leq C_{k,\ell} \qquad (V > 0). \end{aligned} \tag{107}$$

Section 6 proves this estimate. Write $\mathcal{M} = \mathcal{M}_2$. At even depth $n \geq 8$, (75) restricts t to $t = r\sqrt{N}/2$ with Ramanujan weight $(\sqrt{N}/2)(-1)^r$. On the other hand, at odd depth $n \geq 7$, (76) restricts t to $t = r\sqrt{N/2}$ with weight $c_{2^{(n+1)/2}}(t)/\sqrt{2} = (\sqrt{N}/2)(-1)^r$. In both cases the argument of the inverse sine is $r/(4\sqrt{P/N})$, and the discriminant is negative exactly when $P > Nr^2/16$.

All supported t are divisible by 8. The even-depth coefficient is $\mathcal{G}_t/2$ by (90), while the odd-depth coefficient is $\mathcal{G}_t$ by Theorem 5.6. Since t is r multiplied by a power of 2, $\mathcal{G}_t = \mathcal{G}_r$, and both give

$$\frac{\sqrt{Nu}}{2\pi|H|} \sum_{|r|<4\sqrt{u/N}} (-1)^r \cos\left((k-1)\arcsin\frac{r}{4\sqrt{u/N}}\right) \mathcal{G}_r = c_N \mathcal{M}(u/N), \qquad c_N = \frac{(k-1)N}{32|H|}. \tag{108}$$

Here (84) gives $\mathcal{G}_r = (\pi^2/8)\mathfrak{p}_2(r)$, and

$$\cos((k-1)\arcsin x) = s_k U_{k-2}(x)\sqrt{1-x^2}, \qquad |c_N - 1| \leq C_k N^{-1/2}, \tag{109}$$

by the defining sine formula and Lemma 2.5, respectively.

To sum the prime-average errors over the growing index set, we need a bound on variation with respect to u that is uniform in t. For $X \geq 2$, $h = \widehat{\phi}$, at even depth set

$$\Phi_t(u) = u^{-1/2} h(\log u/L) \cos\left((k-1)\arcsin\frac{t}{2\sqrt{u}}\right), \qquad A_t = \max(X, t^2/4). \tag{110}$$

At odd depth replace $2\sqrt{u}$ by $2\sqrt{2u}$ and $t^2/4$ by $t^2/8$. Keep only $A_t < 2X$. The discriminants are negative for every prime in $(A_t, 2X]$. At a zero-discriminant endpoint, the cosine vanishes and its derivative has an integrable square-root singularity. The angle is monotone with variation at most $\pi/2$, so the cosine factor has variation at most $(k-1)\pi/2$. Differentiating the other factors gives

$$\begin{aligned} \max_{[A_t,2X]} |\Phi_t| + \int_{A_t}^{2X} |\Phi_t'(u)|\, du &\leq C_k(\|h\|_\infty + \|h'\|_\infty) X^{-1/2}, \\ \#\{r : Nr^2/16 < 2X\} &\leq 1 + 8\sqrt{2X/N}. \end{aligned} \tag{111}$$

For even depth, expand $w(P)$ by (90) and apply (104) to each. Their fixed coefficients have total absolute value $9/16$. At odd depth apply (105). The factor outside the sum over trace indices is bounded by $C_k N^{-1/2}$ after division by $|H|$, by (71). Summing (111) therefore proves

$$\begin{aligned} \Big| \sum_{X<P\leq 2X} \frac{\log P}{P} h(\log P/L) \frac{A_P^{\mathrm{ell}}}{|H|} - c_N \int_X^{2X} \mathcal{M}(u/N) h(\log u/L) \frac{du}{u} \Big| \\ \leq C_{k,\phi,\epsilon}\{N^{-1/2} X^{1/4+\epsilon} + N^{-1} X^{3/4+\epsilon}\}. \end{aligned} \tag{112}$$

Here ell indicates only the elliptic part of A_P is included.

The second error comes from the growing trace-index count.

Let $X_s = (k^2N)^s$, where $0 < s < 4/3$ and $\operatorname{supp} h \subset [-s, s]$. The bounds (111)–(112) also hold on a truncated last interval. Use consecutive dyadic intervals beginning at 2, truncating the last if necessary. A geometric series, and (112) with $\epsilon \le \eta/(2(1+s))$, give

$$\begin{aligned}\Big|\frac{2}{L}\sum_{P\ge 3}\frac{\log P}{P}h(\log P/L)\frac{A_P}{|H|} - \frac{2c_N}{L}\int_0^\infty \mathcal{M}(v)h(\log(Nv)/L)\frac{dv}{v}\Big| \\ \le C_{k,\phi,s,\eta}\left\{N^{-1/2} + \frac{N^{-1/2+s/4+\eta} + N^{-1+3s/4+\eta}}{L}\right\}.\end{aligned} \tag{113}$$

The normalized even-depth hyperbolic term is bounded by $C_k N^{-1/2}P^{1-k/2}$, and Chebyshev with partial summation gives

$$\frac{1}{L}\sum_{P\le X_s}\frac{\log P}{P}N^{-1/2}P^{1-k/2} \le C_{k,s}N^{-1/2}. \tag{114}$$

Extending the normalized integral in (113) to $0 < v < 2/N$ introduces an error of absolute value at most $C_{k,\phi}N^{-1/2}/L$, by the first bound of Proposition 5.7.

Integrate the bounded primitive against $h(\log(Nv)/L)$. The boundary terms vanish by compact support, and the derivative integral is $\|h'\|_1$, giving

$$\begin{aligned}\Big|\int_0^\infty \{\mathcal{M}(v) - \tfrac{1}{2}\mathbf{1}_{v\ge 1}\}h(\log(Nv)/L)\frac{dv}{v}\Big| \le C_k\|h'\|_1, \\ \frac{2}{L}\int_0^\infty \mathcal{M}(v)h(\log(Nv)/L)\frac{dv}{v} = \int_1^\infty h(u)\,du + O_{k,\phi}(L^{-1}).\end{aligned} \tag{115}$$

In the second line the integral of the step-function term begins at $\log N/L$, whose distance from 1 is exactly $2\log k/L$. Equations (109) and (115) also justify replacing c_N by 1 with error $C_{k,\phi}N^{-1/2}$.

The explicit formula and the signed average. Average (56) with weight ε_f. Lemma 4.5 and Section 4 give absolute convergence at complex ordinates for the present test function. The term $j = 1$ is the prime expression in (113), with a minus sign. The term independent of the newform contributes at most $C_{k,\phi}N^{-1/2}$ by (59) and (71). For the square term, $a_f(p^2) = \lambda_f(p^2) - 1$. Let $Z = X_s^{1/2}$ and apply the two Chebyshev estimates in (50) to get

$$\begin{aligned}\Big|\frac{2}{L|H|}\sum_f \varepsilon_f \sum_{p\le Z}\frac{\log p}{p}a_f(p^2)h(2\log p/L)\Big| \\ \le C_{k,\phi,s,\eta}\{N^{-1+s/2+\eta}/L + N^{-1/2+\eta}/L + N^{-1/2}\}.\end{aligned} \tag{116}$$

Use Lemma 5.1 with $\delta = \eta/(1+s)$ at the odd square indices, and (71) for $|H|$ and the -1 term.

For $j \ge 3$, use $|a_f(p^j)| \le 2$ and the absolute majorant in (60), which is at most 48. Thus

$$\begin{aligned}\Big|\frac{2}{L|H|}\sum_f \varepsilon_f \sum_{p\ge 3}\sum_{j\ge 3}\frac{\log p}{p^{j/2}}a_f(p^j)h(j\log p/L)\Big| \\ \le \frac{192\|h\|_\infty}{L}.\end{aligned} \tag{117}$$

Theorem 5.8 (Signed one-level density)**.** *Assume the Riemann hypothesis for every primitive Dirichlet L-function, including the ζ function of conductor 1. Let $k \ge 2$ be even and let $n \ge 7$. Then, let ϕ be an even Schwartz function whose Fourier transform $h = \widehat{\phi}$ is supported in*

$(-4/3, 4/3)$. *Choose* $0 < s < 4/3$ *with* $\operatorname{supp} h \subset [-s, s]$, *and let* η *be positive and smaller than both* $1 - 3s/4$ *and* $1/2 - s/4$. *Then*

$$\begin{aligned}\Big|S_N(\phi) + \frac{1}{2}\int_{|u|>1} h(u)\,du\Big| \\ \le C_{k,\phi,s,\eta}\Big\{\frac{1}{L} + N^{-1+3s/4+\eta} + N^{-1/2+s/4+\eta} \\ + N^{-1+s/2+\eta} + N^{-1/2+\eta}\Big\}.\end{aligned} \tag{118}$$

In particular the error is $O_{k,\phi}(L^{-1})$, *and* $S_N(\phi) \to -\frac{1}{2}\int_{|u|>1} h$ *as* $n \to \infty$.

Proof. Combine (113), (115), (56), (116) and (117), and use evenness to obtain $-\int_1^\infty h = -\frac{1}{2}\int_{|u|>1} h$. □

Proof of Corollary 1.3. Set $\delta_N = R_N/d_N$. For $n \ge 7$, by (71) both sign classes are nonempty and $|(1 \pm \delta_N)^{-1} - 1| \le 16(k-1)^{-1}N^{-1/2}$. The indicator $(1 \pm \varepsilon_f)/2$ gives

$$D_N^{\pm}(\phi) = \frac{U_N(\phi) \pm S_N(\phi)}{1 \pm \delta_N}. \tag{119}$$

For $n \ge 8$, use Theorems 1.1 and 1.2. Their main terms remain bounded, which allows the removal of the denominator at cost $O_{k,\phi}(N^{-1/2})$, giving

$$D_N^{\pm}(\phi) = \widehat{\phi}(0) + \frac{\phi(0)}{2} \mp \frac{1}{2}\int_{|\xi|>1} \widehat{\phi}(\xi)\,d\xi + O_{k,\phi}(L^{-1}).$$

Using $\phi(0) = \int_{\mathbb{R}} \widehat{\phi}$ identifies the two main terms with the densities in Section 1.3. □

6. The murmuration density and its shape

For the density (106), we compute the logarithmic mean needed in the signed one-level formula and determine its pointwise oscillations. Throughout, k is fixed and even, $s_k = (-1)^{k/2+1}$, and $m = k - 1$, while F_k and $\mathfrak{p}_\ell$ are as in (106).

Theorem 6.1. *Let* ℓ *be a prime and* $\mathcal{M}_\ell$ *as in* (106). *Then*

$$\begin{aligned}&\mathcal{M}_\ell(y) = \frac{2\pi}{k-1}\sqrt{y} \quad (0 < y \le 1/(4\ell^2)),\\ &\Big|\int_1^Y \mathcal{M}_\ell(y)\frac{dy}{y} - \frac{1}{2}\log Y\Big| \le C_{k,\ell} \quad (Y \ge 1),\\ &\mathcal{M}_\ell(y) - \frac{1}{2} = y^{1/4}\mathcal{A}_{k,\ell}(\sqrt{y}) + O_{k,\ell}(1) \quad (y \ge 1),\end{aligned} \tag{120}$$

where $\mathcal{A}_{k,\ell}$ *is the absolutely and uniformly convergent trigonometric series* (131) *below. The density is unbounded above and below, with oscillations attaining order* $y^{1/4}$, *and does not converge.*

Proof of the first line of (120). For $0 < y \le 1/(4\ell^2)$, only $r = 0$ satisfies $|r| < 2\ell\sqrt{y}$. Its weight $1 - 1/\ell$, $\mathfrak{p}_\ell(0) = 1$, and $F_k(0) = s_k$ give the stated value. □

To separate the arithmetic weight from the kernel before applying Poisson summation, write

$$\begin{aligned} B_\ell &= \prod_{p\neq\ell}\left(1-\frac{1}{p(p-1)}\right),\\ b_\ell(d) &= \mu(d)^2\mathbf{1}_{(d,\ell)=1}\prod_{p\mid d}(p^2-p-1)^{-1}, \qquad c_\ell(d)=db_\ell(d),\\ \mathfrak{p}_\ell(r) &= B_\ell\sum_{d\mid r}b_\ell(d). \end{aligned} \tag{121}$$

At $r=0$, the divisor sum runs over all $d\geq 1$ and equals B_ℓ^{-1} by an absolutely convergent Euler product.

Lemma 6.2. *For* $\operatorname{Re} z>0$,

$$\begin{aligned} \sum_{d\geq 1}c_\ell(d)d^{-z} &= \zeta(1+z)H_\ell(z),\\ H_\ell(z) &= (1-\ell^{-1-z})\prod_{p\neq\ell}\left(1+\frac{p^{1-z}}{p^2-p-1}\right)(1-p^{-1-z}) = \sum_{a\geq 1}h_\ell(a)a^{-z}. \end{aligned} \tag{122}$$

The last series converges absolutely for $\operatorname{Re} z>-1/2$*; in particular* $\sum_a|h_\ell(a)|a^{3/8}\leq C_\ell$.

Proof. Factoring the Euler product of c_ℓ against $\zeta(1+z)$ leaves coefficients of p^{-z} and p^{-2z} equal to $p/(p^2-p-1)-1/p$ and $-1/(p^2-p-1)$, both bounded in modulus by $C_{\mathrm{abs}}p^{-2}$. Thus $\sum_p p^{-2+2\sigma}<\infty$ for $\sigma<1/2$ proves the assertion. □

Lemma 6.3.

$$\begin{aligned} \sum_{d\leq X}c_\ell(d) &= R_\ell\log X + C_\ell^0 + O_\ell(X^{-1/4}) \quad (X\geq 1),\\ R_\ell &= H_\ell(0), \qquad C_\ell^0 = \gamma R_\ell - \sum_a h_\ell(a)\log a,\\ B_\ell R_\ell &= (1-\ell^{-1})\prod_{p\neq\ell}(1-p^{-2}) = \frac{\ell}{(\ell+1)\zeta(2)}. \end{aligned} \tag{123}$$

Proof. Equation (122) gives $c_\ell(d)=\sum_{ab=d}h_\ell(a)/b$. Thus

$$\sum_{d\leq X}c_\ell(d) = \sum_{a\leq X}h_\ell(a)\sum_{b\leq X/a}\frac{1}{b}, \qquad \left|\sum_{b\leq T}\frac{1}{b}-\log T-\gamma\right|\leq C_{\mathrm{abs}}T^{-1} \quad (T\geq 1). \tag{124}$$

The harmonic-sum bound follows by integral comparison. By the bound on $\sum_a|h_\ell(a)|a^{3/8}$ in Lemma 6.2 and $\log u\leq 8u^{1/8}$ for $u\geq 1$, substituting $T=X/a$ and extending the main sums to infinity costs at most $C_\ell X^{-1/4}$. Multiplication of the local factors gives $B_\ell R_\ell$. □

Equation (123) bounds the mass on each $[X,2X]$ by C_ℓ, so dyadic summation gives

$$\begin{aligned} &\sum_{d\leq X}c_\ell(d)d^{1/2}\leq C_\ell X^{1/2},\\ &\sum_{d>X}c_\ell(d)d^{-1/2}\leq C_\ell X^{-1/2},\\ &\sum_{d>X}c_\ell(d)d^{-1}\leq C_\ell X^{-1}, \qquad \sum_{d\geq 1}b_\ell(d)\sqrt{d}\leq C_\ell \quad (X\geq 1). \end{aligned} \tag{125}$$

For the positive odd integer $m = k-1$, the Bessel integral, recurrence, asymptotic, and odd-order antiderivative [DLM, (10.9.2), (10.6.1), (10.17.3), (10.22.9)] give

$$\begin{aligned} J_m(t) &= \sqrt{\frac{2}{\pi t}}\cos\!\left(t - \frac{(2m+1)\pi}{4}\right) + O_k(t^{-3/2}) \quad (t \ge 1), \\ |J_m(t)| &\le C_k \min(t^m, t^{-1/2}) \quad (t > 0), \\ I_m(T) := \int_0^T J_m(t)\,dt &= 1 - J_0(T) - 2\sum_{j=1}^{(m-1)/2} J_{2j}(T), \\ |I_m(T)| &\le C_k \quad (T \ge 0), \qquad |I_m(T) - 1| \le C_k T^{-1/2} \quad (T \ge 1). \end{aligned} \tag{126}$$

The defining power series [DLM, (10.2.2)] gives the small-argument bound on $0 < t \le 1$.

The substitution $u = \cos\theta$ and identity $2\sin\theta\sin((k-1)\theta) = \cos((k-2)\theta) - \cos(k\theta)$ in the Fourier integral, followed by the Bessel integral and recurrence, give

$$\widehat{F}_k(\xi) = \frac{(k-1)s_k}{2\xi} J_{k-1}(2\pi\xi) \quad (\xi \ne 0), \qquad |\widehat{F}_k(\xi)| \le C_k(1+|\xi|)^{-3/2}. \tag{127}$$

Parity treats $\xi < 0$, continuity gives the value at zero, and (126) gives the decay.

Poisson summation works even when a lattice point meets the boundary of the kernel's support. The continuous periodization of F_k has the absolutely summable Fourier coefficients (127). Its Fejér means converge uniformly, since the nonnegative kernels $J^{-1}|\sum_{j=0}^{J-1} e^{2\pi i j x}|^2$ have integral 1 and are bounded by $1/(J\sin^2(\pi\delta))$ outside distance δ from the integers. They therefore identify the periodization with its uniformly convergent Fourier series.

Lemma 6.4. *Let g_ℓ be (106) with $\mathfrak{p}_\ell(r)$ replaced by 1. Then*

$$\begin{aligned} g_\ell(y) &= \frac{2\pi\sqrt{y}}{1-1/\ell}\sum_{\nu\ge1}\left\{\frac{J_m(4\pi\nu\sqrt{y})}{\nu} - \frac{J_m(4\pi\ell\nu\sqrt{y})}{\ell\nu}\right\}, \\ \mathcal{M}_\ell(y) &= B_\ell \sum_{d\ge1} c_\ell(d) g_\ell(y/d^2), \\ |g_\ell(y)| &\le C_{k,\ell} y^{1/4} \quad (y \ge 1), \qquad |g_\ell(y)| \le C_{k,\ell} y^{1/2} \quad (0 < y \le 1). \end{aligned} \tag{128}$$

Proof. Poisson summation on the full lattice and its ℓ-sublattice cancels the zero frequencies, and (127) gives the remaining Bessel series. Insert (121) and use $(d,\ell) = 1$ to get the divisor sum. Its tail converges absolutely. For $d > 2\ell\sqrt{y}$, only the zero lattice index contributes, giving $g_\ell(y/d^2) = 2\pi\sqrt{y}/((k-1)d)$. The same observation gives the small-y bound, with finitely many indices on the remaining compact interval. For large y, use (126) and $\sum_\nu \nu^{-3/2} < \infty$. □

Lemma 6.5. *For $G_\ell(Z) = \int_0^Z g_\ell(y)dy/y$ and $x = \sqrt{Z}$,*

$$\begin{aligned} G_\ell(Z) &= \frac{1}{1-1/\ell}\sum_{\nu\ge1}\nu^{-2}\{I_m(4\pi\nu x) - \ell^{-2} I_m(4\pi\ell\nu x)\}, \\ G_\ell(\infty) &= (1+1/\ell)\zeta(2), \\ |G_\ell(Z) - G_\ell(\infty)| &\le C_{k,\ell} Z^{-1/4} \quad (Z \ge 1), \\ |G_\ell(Z)| &\le C_{k,\ell} Z^{1/2} \quad (0 < Z \le 1). \end{aligned} \tag{129}$$

Proof. By (126), the absolute integral of the νth term is at most $C_{k,\ell}\nu^{-3/2}Z^{1/4}$, which permits termwise integration. Boundedness of I_m allows passage to infinity, where $I_m \to 1$. Its remainder in (126) gives the large-Z bound. Integrating $|g_\ell(y)| \le C_{k,\ell}\sqrt{y}$ gives the small-Z bound. □

Proof of the logarithmic-mean assertion in Theorem 6.1. Set $x = \sqrt{Y}$ for $Y \geq 1$. Integrating (128) gives $B_\ell \sum_d c_\ell(d)\{G_\ell(Y/d^2) - G_\ell(1/d^2)\}$. The absolute tail in Lemma 6.4 justifies the interchange, and the terms $G_\ell(1/d^2)$ have a bounded total by (125). For $d \leq x$, replace $G_\ell(Y/d^2)$ by $G_\ell(\infty)$. For $d > x$, use the small-argument bound in (129). The errors are at most $C_{k,\ell}$ times $x^{-1/2}\sum_{d\leq x} c_\ell(d)d^{1/2}$ and $x\sum_{d>x} c_\ell(d)/d$, both bounded by (125). Equation (123) therefore gives

$$\int_1^Y \mathcal{M}_\ell(y)\frac{dy}{y} = B_\ell R_\ell G_\ell(\infty)\log x + O_{k,\ell}(1) = \frac{1}{2}\log Y + O_{k,\ell}(1), \tag{130}$$

since (123) and (129) give $B_\ell R_\ell G_\ell(\infty) = 1$. □

Proof of Proposition 5.7. For $V \geq 1$, apply (130). On $(0,1]$, (128) gives

$$|\mathcal{M}_\ell(v)| \leq C_{k,\ell} B_\ell \sqrt{v} \sum_d b_\ell(d) \leq C_{k,\ell}\sqrt{v},$$

and hence $\int_0^1 |\mathcal{M}_\ell(v)|dv/v \leq C_{k,\ell}$. □

For pointwise behavior, retain the leading Bessel terms rather than integrating them. Set $\delta_k = (2k-1)\pi/4$ and

$$\mathcal{A}_{k,\ell}(x) = \frac{\sqrt{2}B_\ell}{1-1/\ell}\sum_{d\geq 1} b_\ell(d)\sqrt{d}\sum_{\nu\geq 1}\nu^{-3/2}\big\{\cos(4\pi\nu x/d - \delta_k) - \ell^{-3/2}\cos(4\pi\ell\nu x/d - \delta_k)\big\}. \tag{131}$$

This series converges absolutely and uniformly by (125), and hence defines a bounded continuous function.

Proof of the oscillation assertion in Theorem 6.1. Apply (126) to the $d \leq \sqrt{y}$ terms in (128). The accumulated Bessel remainder, the original tail, and the leading-term tail are bounded by $C_{k,\ell}$ times the following three quantities, respectively:

$$\begin{aligned} y^{-1/4}\sum_{d\leq\sqrt{y}} c_\ell(d)d^{1/2} &\leq C_\ell, \\ \sqrt{y}\sum_{d>\sqrt{y}} \frac{c_\ell(d)}{d} &\leq C_\ell, \\ y^{1/4}\sum_{d>\sqrt{y}} c_\ell(d)d^{-1/2} &\leq C_\ell. \end{aligned} \tag{132}$$

By (125), these give the expansion in (120). Subtracting $1/2$ changes only the bounded remainder. Boundedness of $\mathcal{A}_{k,\ell}$ also gives

$$|\mathcal{M}_\ell(y)| \leq C_{k,\ell} y^{1/4} \quad (y \geq 1). \tag{133}$$

All frequencies in (131) are nonzero, so uniform convergence gives mean zero for $\mathcal{A}_{k,\ell}$. Its coefficient at frequency 2 is

$$\lim_{T\to\infty}\frac{1}{T}\int_0^T \mathcal{A}_{k,\ell}(x)e^{-4\pi i x}\,dx = \frac{B_\ell e^{-i\delta_k}}{\sqrt{2}(1-1/\ell)}\sum_{d\geq 1}\frac{b_\ell(d)}{d} \neq 0. \tag{134}$$

The first cosine in (131) has frequency 2 precisely when $\nu = d$, but the second would require $d = \ell\nu$, which is excluded by $(d,\ell) = 1$ on the support of b_ℓ. The coefficient is nonzero since $b_\ell \geq 0$ and $b_\ell(1) = 1$.

To obtain both signs quantitatively, set $K_\ell = \sqrt{2}B_\ell/(1-1/\ell)$, $S_\ell = \sum b_\ell(d)/d$, $T_\ell = \sum b_\ell(d)\sqrt{d}$, $q = K_\ell S_\ell/2$, and $Q = K_\ell(1+\ell^{-3/2})\zeta(3/2)T_\ell$. Then $|\mathcal{A}| \leq Q$, while its two conjugate coefficients

in (134) give mean square at least $2q^2$ by finite orthogonality and uniform approximation. Since its mean is zero, each of its positive and negative parts has mean at least q^2/Q. Finite approximations to (131) have rational frequencies. For any prescribed error, choose a sufficiently close finite approximation. Integer multiples of its common period then change $\mathcal{A}$ by at most that error, uniformly in its argument. Hence both signs recur arbitrarily far out, and (120) gives sequences tending to infinity with

$$\mathcal{M}_\ell(y_j^+) \geq c_{k,\ell}(y_j^+)^{1/4}, \qquad \mathcal{M}_\ell(y_j^-) \leq -c_{k,\ell}(y_j^-)^{1/4}, \qquad c_{k,\ell} > 0. \tag{135}$$

One may take $c_{k,\ell} = q^2/(4Q)$ after discarding finitely many points.

□

7. Support limitations and further directions

The thresholds $3/2$ and $4/3$ arise from the following estimates. They do not need to be optimal for the density statements.

The unconditional support $3/2$. For $a > 0$, a smooth function g supported in $[0, a]$, and $X = (k^2N)^a$, let

$$\mathcal{T}_N(g) = \frac{1}{L} \sum_{P \leq X} \frac{\log P}{\sqrt{P}} E_N(P) g\left(\frac{\log P}{L}\right). \tag{136}$$

Combining (35) and (44) as in (45) gives, for $n \geq 8$ and $0 < \delta < 1/2$,

$$|\mathcal{T}_N(g)| \leq \frac{48aK_\delta\sqrt{2}}{1 - 2^{-1-\delta}} \|g\|_\infty N^{-3/2} X^{1+\delta}. \tag{137}$$

The exponent $-3/2 + a(1 + \delta)$ is negative for $0 < \delta < \min\{1/2, (3/2 - a)/a\}$ when $a < 3/2$, and positive for every $\delta > 0$ when $a \geq 3/2$. The prime powers in Section 4 impose no further restriction.

For $q = N/16$, the underlying integer pair count satisfies

$$2B^{3/2}N^{-1} \leq \sum_{\substack{B < m \leq 2B \\ m \text{ odd}}} T_N(m) \leq 128B^{3/2}N^{-1} \qquad (B \geq \max\{2q, 64\}). \tag{138}$$

For the lower bound, choose the at least $\sqrt{B}/4$ odd integers r such that $|r| \leq \sqrt{B}/2$. For each, the interval $(B - r^2)/q < v \leq (2B - r^2)/q$ contains at least $B/(2q)$ positive integers, and each gives an odd $m = r^2 + qv$ counted by $T_N(m)$. For the upper bound there are at most $4\sqrt{B}$ choices of r with $r^2 < 2B$ and at most $B/q + 1 \leq 3B/(2q)$ choices of v for each. Their product is $96B^{3/2}/N$.

For fixed $0 < a < 2$, each prime $P \leq X$ has at most two surviving indices once the depth is large. Let

$$n_0(k, a) = \max\left\{8, 1 + \left\lceil \frac{\log_2(64k^a)}{1 - a/2} \right\rceil\right\}. \tag{139}$$

If $n \geq n_0(k, a)$ then $\sqrt{X} < q/4$. By (33), a contributing $t = 2r$ has r odd and $r^2 \equiv P \pmod q$. Any two such roots satisfy $r \equiv \pm s \pmod{q/2}$ and within $(-q/4, q/4)$, this forces $r = \pm s$. Hence

$$\#\{r : |r| < \sqrt{P},\ r^2 \equiv P \pmod q\} \leq 2 \qquad (P \leq X,\ n \geq n_0(k, a)). \tag{140}$$

Negating the local matrix leaves its projective fixed lines unchanged, and U_{k-2} is even so these two contributions add. At $k = 2$, $U_0 = 1$ also removes angular oscillation. Extending this argument beyond $3/2$ therefore requires a bound for (136) that improves on its absolute majorant.

The conditional support 4/3**.** Under Dirichlet GRH, the exponent 3/4 in Theorems 5.5 and 5.6 balances the smoothing error $C_\eta M B^{1+\eta} x^{-1/2}$ against the error from the character-sum estimates $C_\eta(M+V)B^{1/2+\eta}x^{1/2+\eta}$ in (103).

The variation bound $M+V \le C_{k,\phi}B^{-1/2}$, the count of at most $1+8\sqrt{2B/N}$ trace indices in (111), and the factor $C_k N^{-1/2}$ outside the trace-index sum give the block error

$$\frac{C_{k,\phi,\epsilon}}{L}\left(N^{-1/2}B^{1/4+\epsilon}+N^{-1}B^{3/4+\epsilon}\right). \tag{141}$$

At $B=(k^2N)^a$, the second term requires $a<4/3$, while the first and (116) permit every fixed $a<2$. A bound with $B^{1/2+\epsilon}$ in both prime-sum theorems, with the same uniformity in the index and test weight, would instead give $C_{k,\phi,\epsilon}L^{-1}(N^{-1/2}B^{\epsilon}+N^{-1}B^{1/2+\epsilon})$ and would yield every fixed support below 2 for S_N. For comparison, Iwaniec–Luo–Sarnak [ILS00, Theorem 1.1] obtain support below 2 at squarefree levels under their stated Riemann hypotheses.

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